

FACULTÉ DES SCIENCES ET TECHNOLOGIES(FST)

TD N°1-Administration des systèmes informatiques 2

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Niveau: L4

Professeur : Ismael SAINT AMOUR

Préparé le 20/10/2026

TD1 Installation et infrastructure Windows Server 2019

Objectifs:

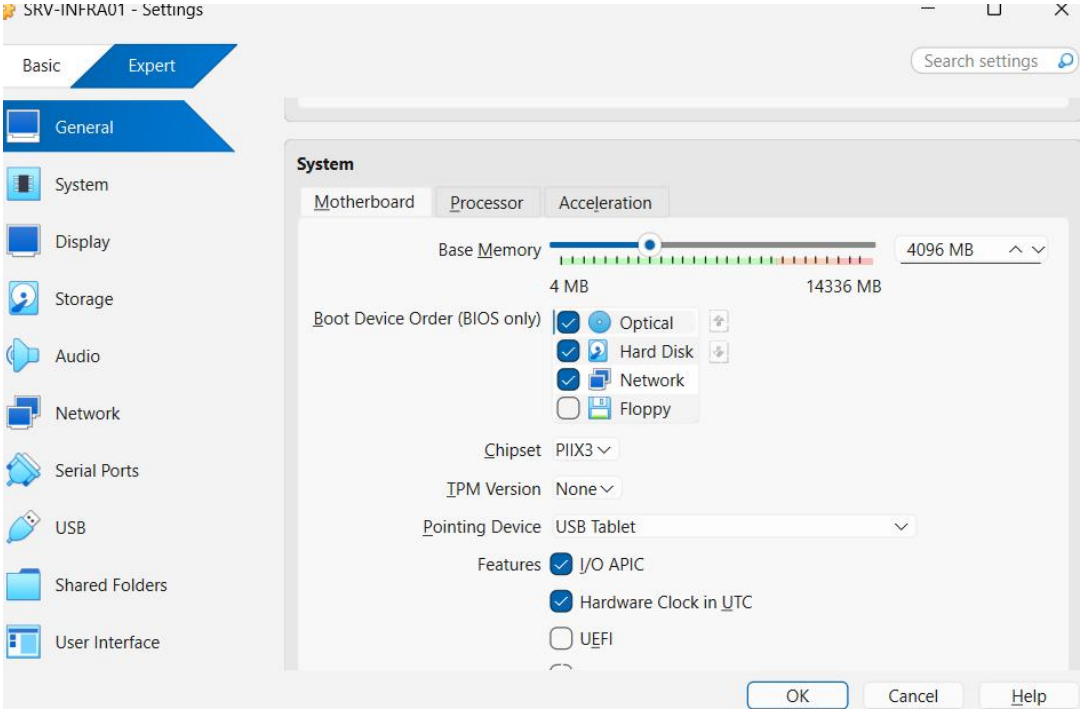
Installer Windows Server 2019 ou 2012 avec une interface graphique.

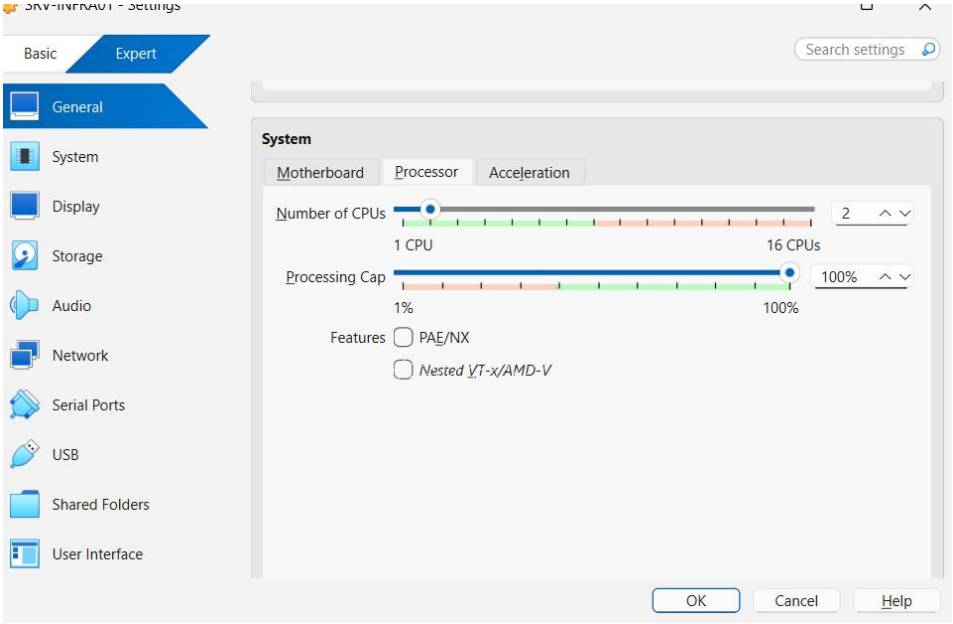
Configurer le nom du serveur et son réseau.

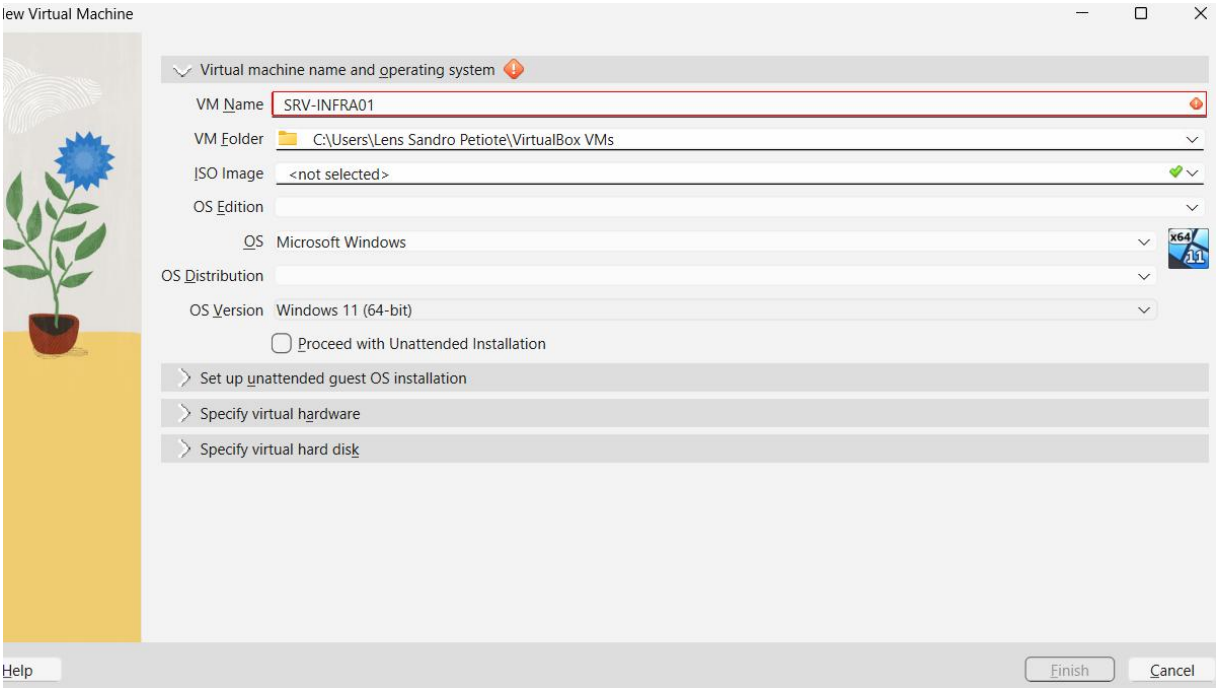
Installer et vérifier des rôles de base.

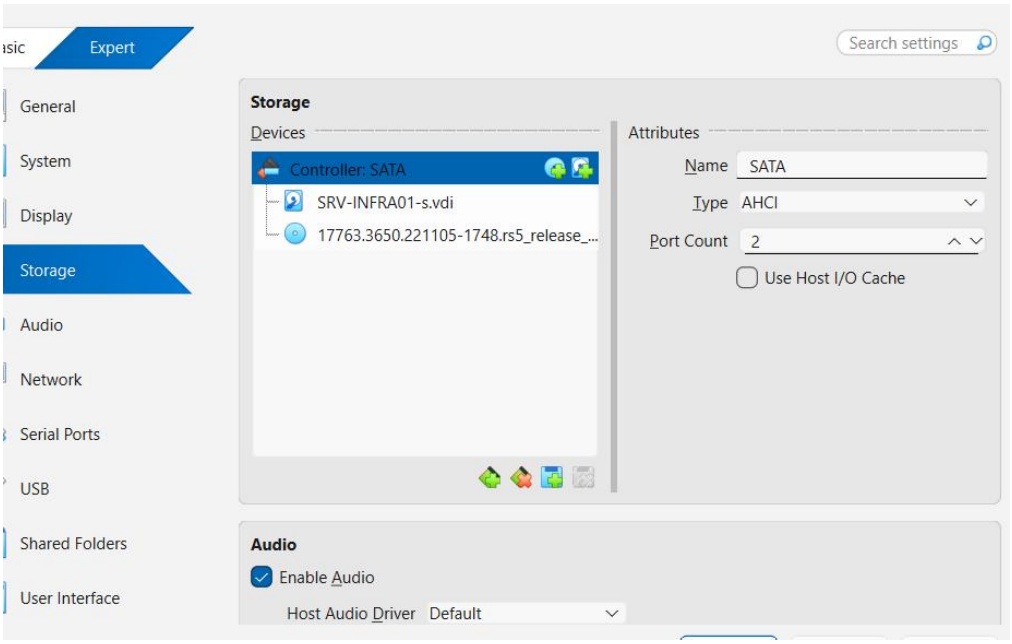
Administrer le serveur avec Server Manager, Windows Admin Center et PowerShell.

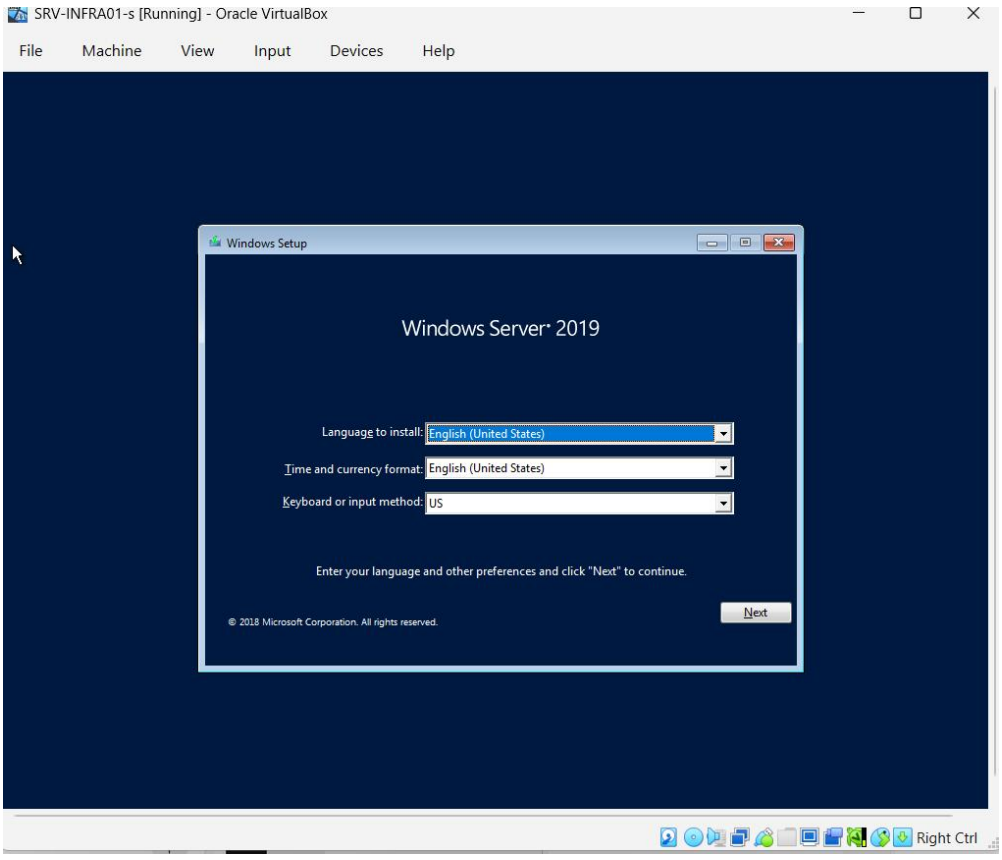
1- Installation de Windows Server 2019

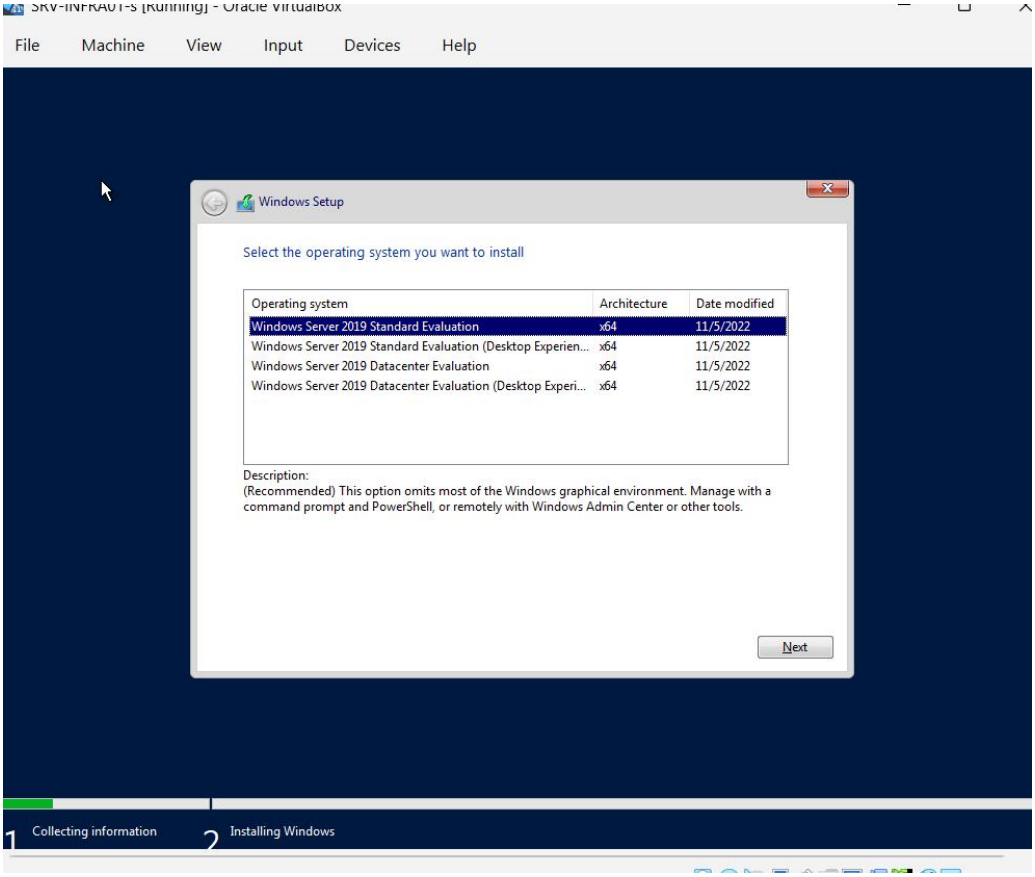


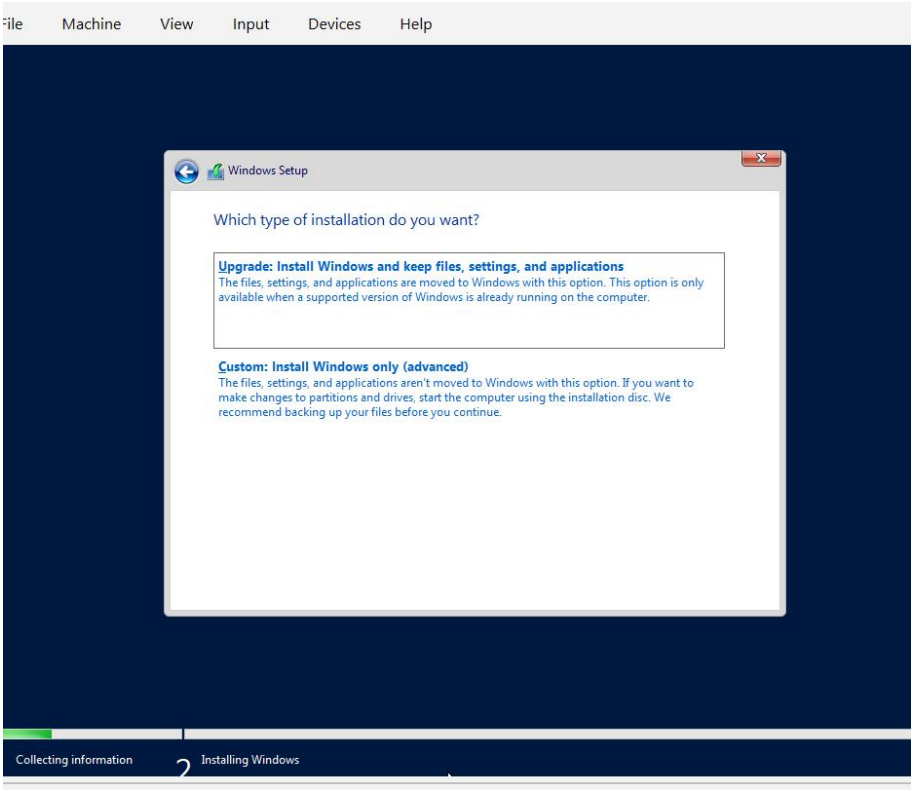

Basic
Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
System
Motherboard
Processor
Acceleration
Number of CPUs
2
1 CPU
16 CPUs
Processing Cap
100%
1%
100%
Features
PAE/NX
Nested VT-x/AMD-V
OK
Cancel
Help

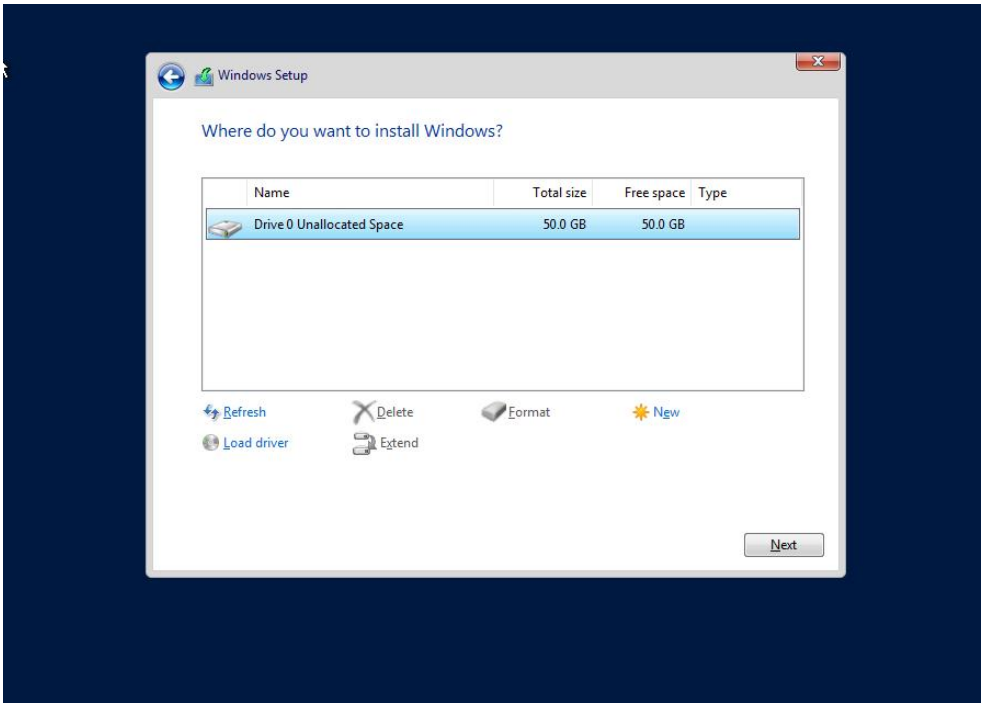

lew Virtual Machine
Virtual machine name and operating system
VM Name
SRV-INFRA01
VM Folder
C:\Users\Lens Sandro Petiote\VirtualBox VMs
ISO Image
<not selected>
OS Edition
OS
Microsoft Windows
OS Distribution
OS Version
Windows 11 (64-bit)
Proceed with Unattended Installation
Set up unattended guest OS installation
Specify virtual hardware
Specify virtual hard disk
Help
Finish
Cancel

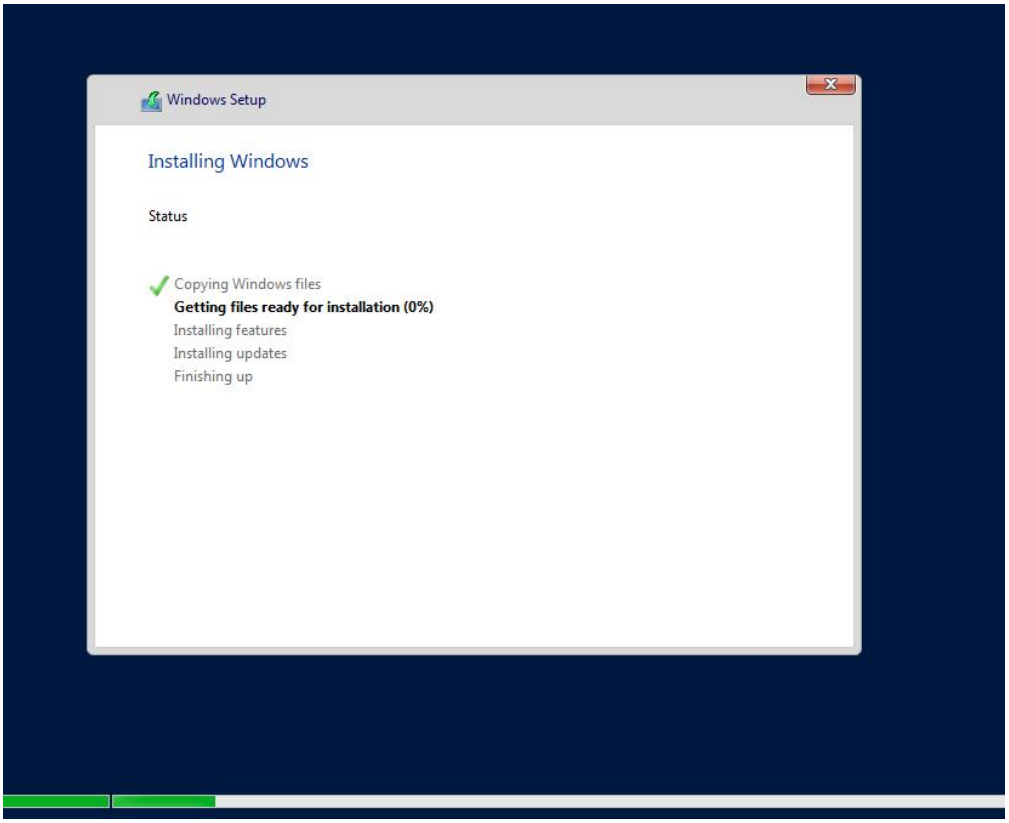

Expert
Search settings
General
System
Display
Storage
Audio
Network
Serial Ports
USB
Shared Folders
User Interface
Storage
Devices
Controller: SATA
SRV-INFRA01-s.vdi
17763.3650.221105-1748.rs5_release_...
Attributes
Name SATA
Type AHCI
Port Count 2
Use Host I/O Cache
Audio
Enable Audio
Host Audio Driver Default


SRV-INFRA01-s [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Windows Setup
Windows Server 2019
Language to install: English (United States)
Time and currency format: English (United States)
Keyboard or input method: US
Enter your language and other preferences and click "Next" to continue.
© 2018 Microsoft Corporation. All rights reserved.
Next
Right Ctrl

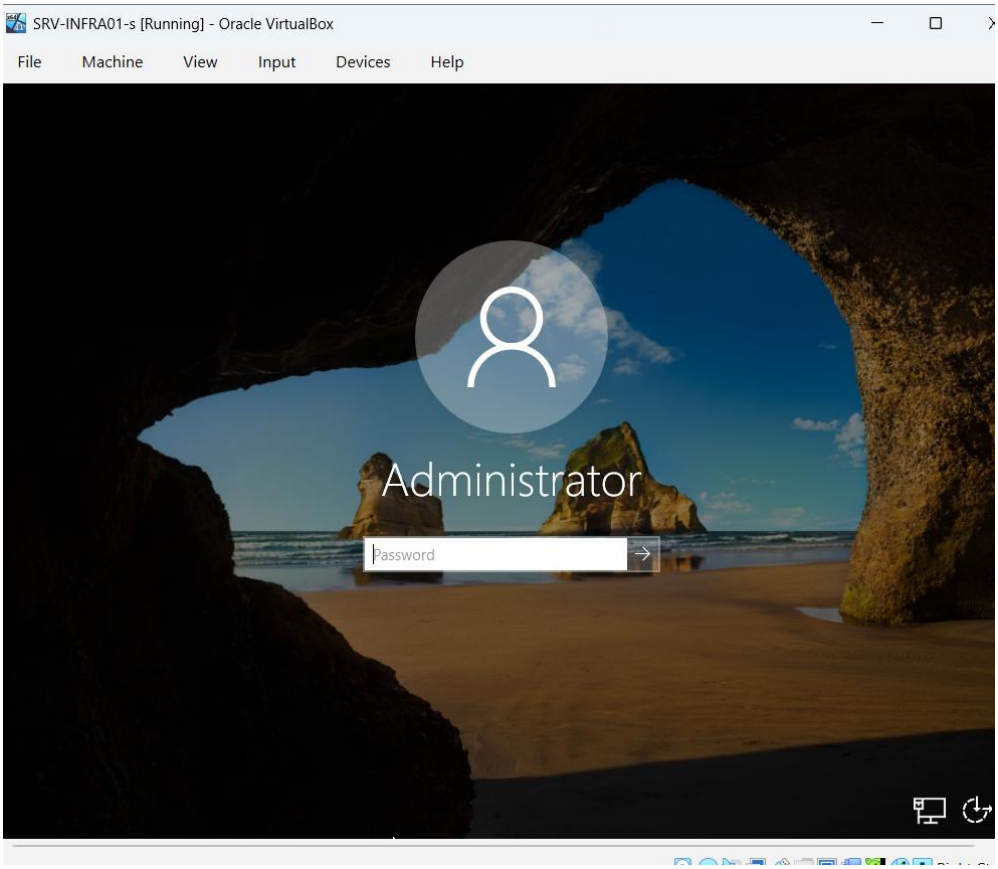

File
Machine
View
Input
Devices
Help
Windows Setup
Select the operating system you want to install
Operating system
Architecture
Date modified
Windows Server 2019 Standard Evaluation
x64
11/5/2022
Windows Server 2019 Standard Evaluation (Desktop Experien...
x64
11/5/2022
Windows Server 2019 Datacenter Evaluation
x64
11/5/2022
Windows Server 2019 Datacenter Evaluation (Desktop Experi...
x64
11/5/2022
Description:
(Recommended) This option omits most of the Windows graphical environment. Manage with a command prompt and PowerShell, or remotely with Windows Admin Center or other tools.
Next
1 Collecting information
2 Installing Windows

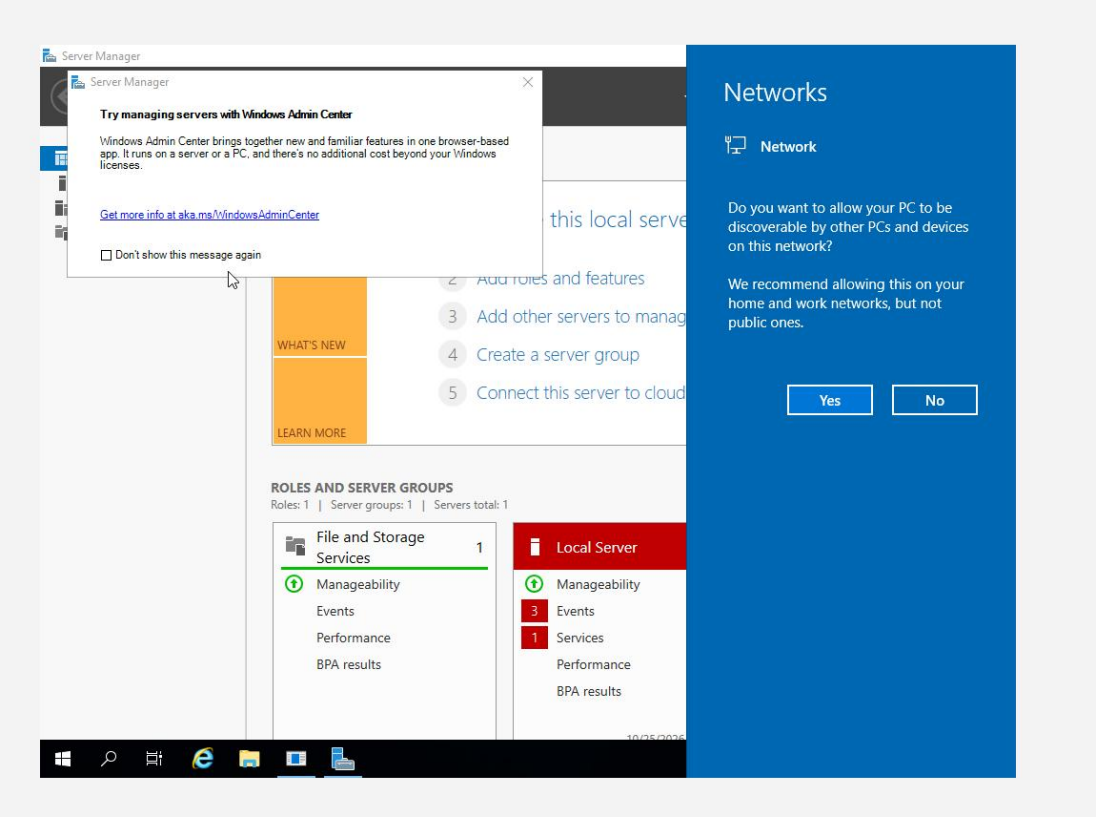

File
Machine
View
Input
Devices
Help
Windows Setup
Which type of installation do you want?
Upgrade: Install Windows and keep files, settings, and applications
The files, settings, and applications are moved to Windows with this option. This option is only available when a supported version of Windows is already running on the computer.
Custom: Install Windows only (advanced)
The files, settings, and applications aren't moved to Windows with this option. If you want to make changes to partitions and drives, start the computer using the installation disc. We recommend backing up your files before you continue.
Collecting information
2 Installing Windows


Windows Setup
Where do you want to install Windows?
Name
Total size
Free space
Type
Drive 0 Unallocated Space
50.0 GB
50.0 GB
Refresh
Delete
Format
New
Load driver
Extend
Next


Windows Setup
Installing Windows
Status
Copying Windows files
Getting files ready for installation (0%)
Installing features
Installing updates
Finishing up


Press Ctrl+Alt+Delete to unlock.
2:51
Sunday, October 25


SRV-INFRA01-s [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Administrator
Password



Questions#1

1.Quelle différence existe-t-il entre Server Core et l'installation avec Expérience utilisateur ?

Server Core est une version de Windows Server qui s'installe sans interface graphique complète. L'administration se fait principalement à l'aide de commandes PowerShell ou d'outils d'administration à distance. Elle consomme moins de ressources et offre une meilleure sécurité grâce à une surface d'attaque réduite.

Tandis que l'installation avec Expérience utilisateur (Desktop Experience) inclut une interface graphique complète similaire à celle de Windows classique. Elle est plus facile à utiliser pour les débutants, mais nécessite davantage de ressources système.

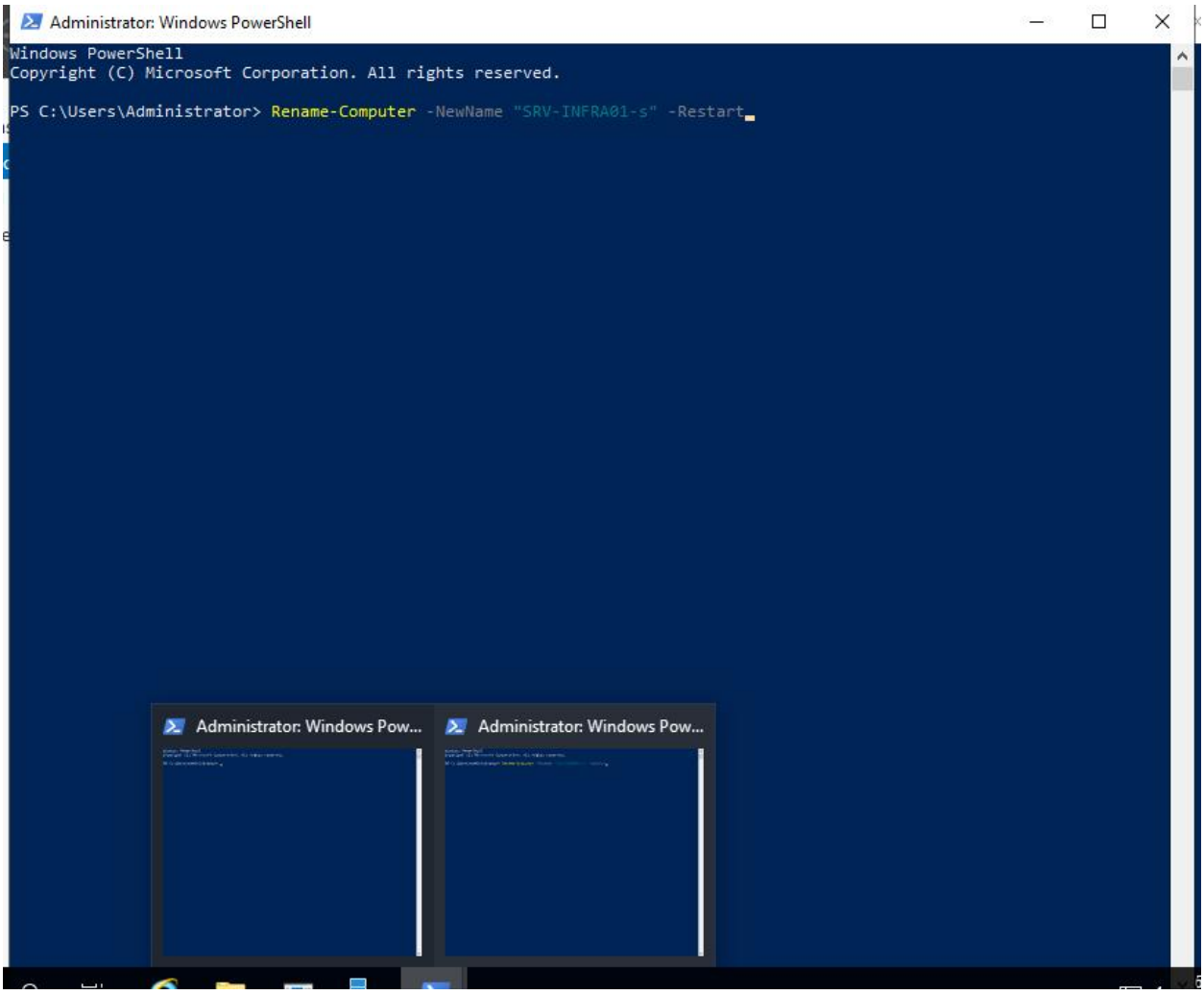
2. Pourquoi utiliser une machine virtuelle pour ce TD ?

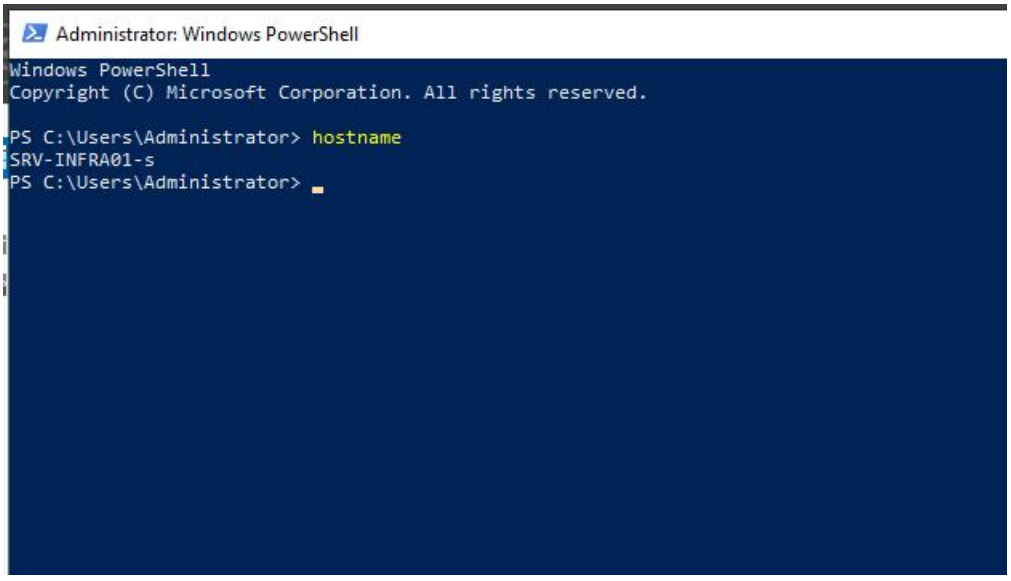
Dans ce TD, une machine virtuelle est utilisée afin d'installer et de configurer Windows Server sans affecter le système d'exploitation principal de l'ordinateur. Elle permet de créer un environnement de test sécurisé où l'on peut réaliser différentes manipulations, comme l'installation du serveur, la configuration des rôles et des services, tout en évitant les risques pour le matériel ou les données de l'ordinateur hôte. Elle offre également la possibilité de recommencer facilement les exercices en cas d'erreur.

3. Quel est le rôle du compte Administrateur ?

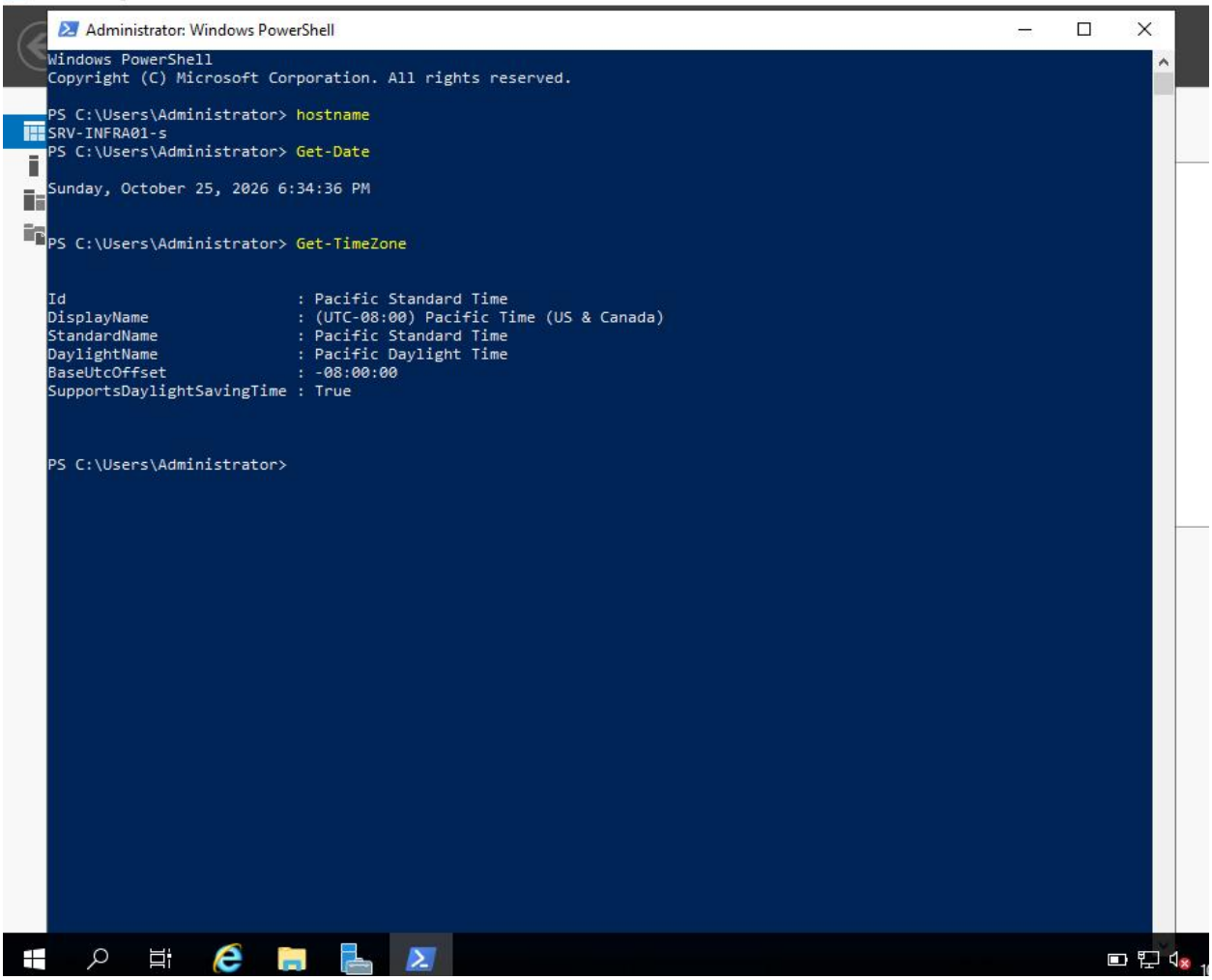
Le compte Administrateur est le compte qui possède tous les droits sur le serveur. Il permet de configurer Windows Server, de gérer les paramètres du système, d'ajouter ou de supprimer des rôles et fonctionnalités, ainsi que de créer et administrer les comptes utilisateurs. Il est indispensable pour réaliser les tâches de configuration et d'administration demandées durant les travaux pratiques.

1- Configuration initiale du serveur

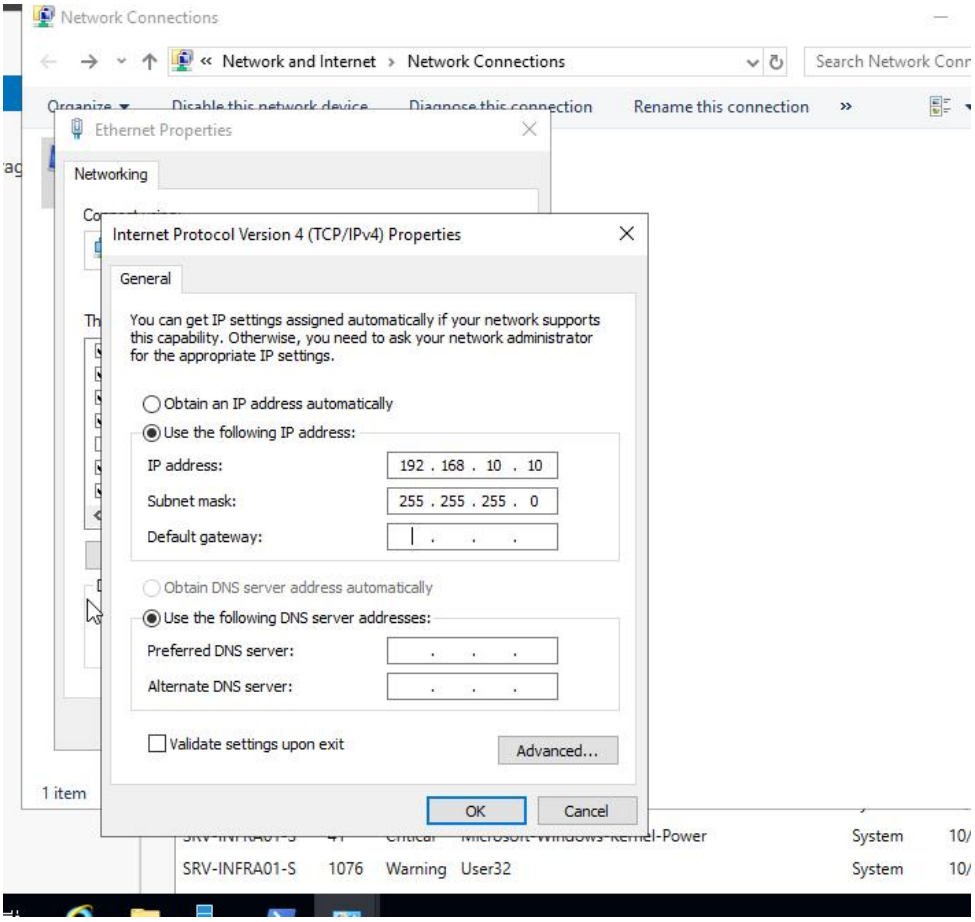


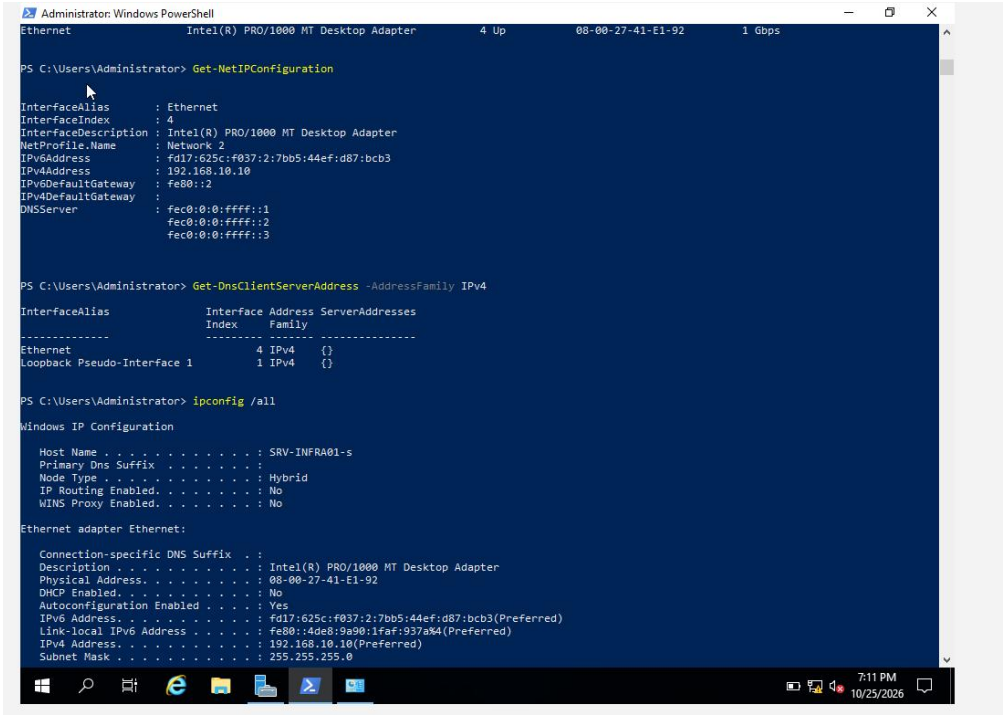


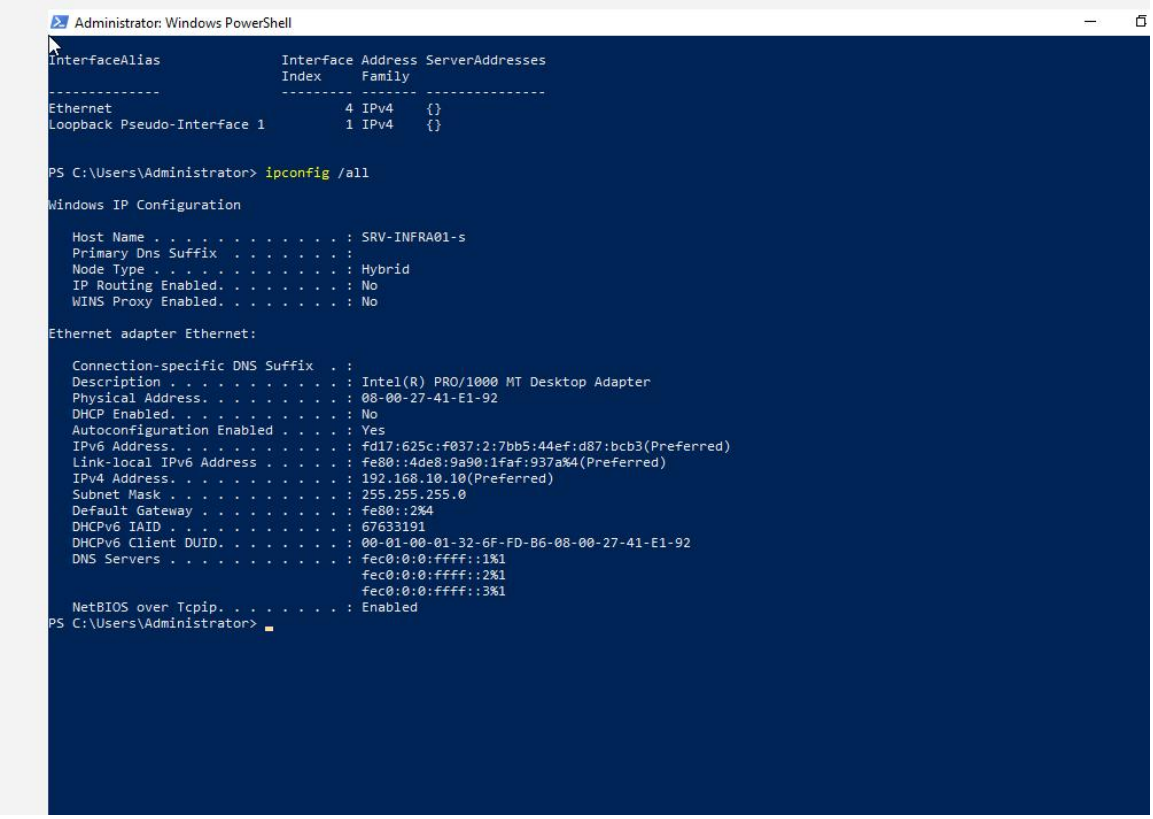
2-Vérifier la date et l'heure



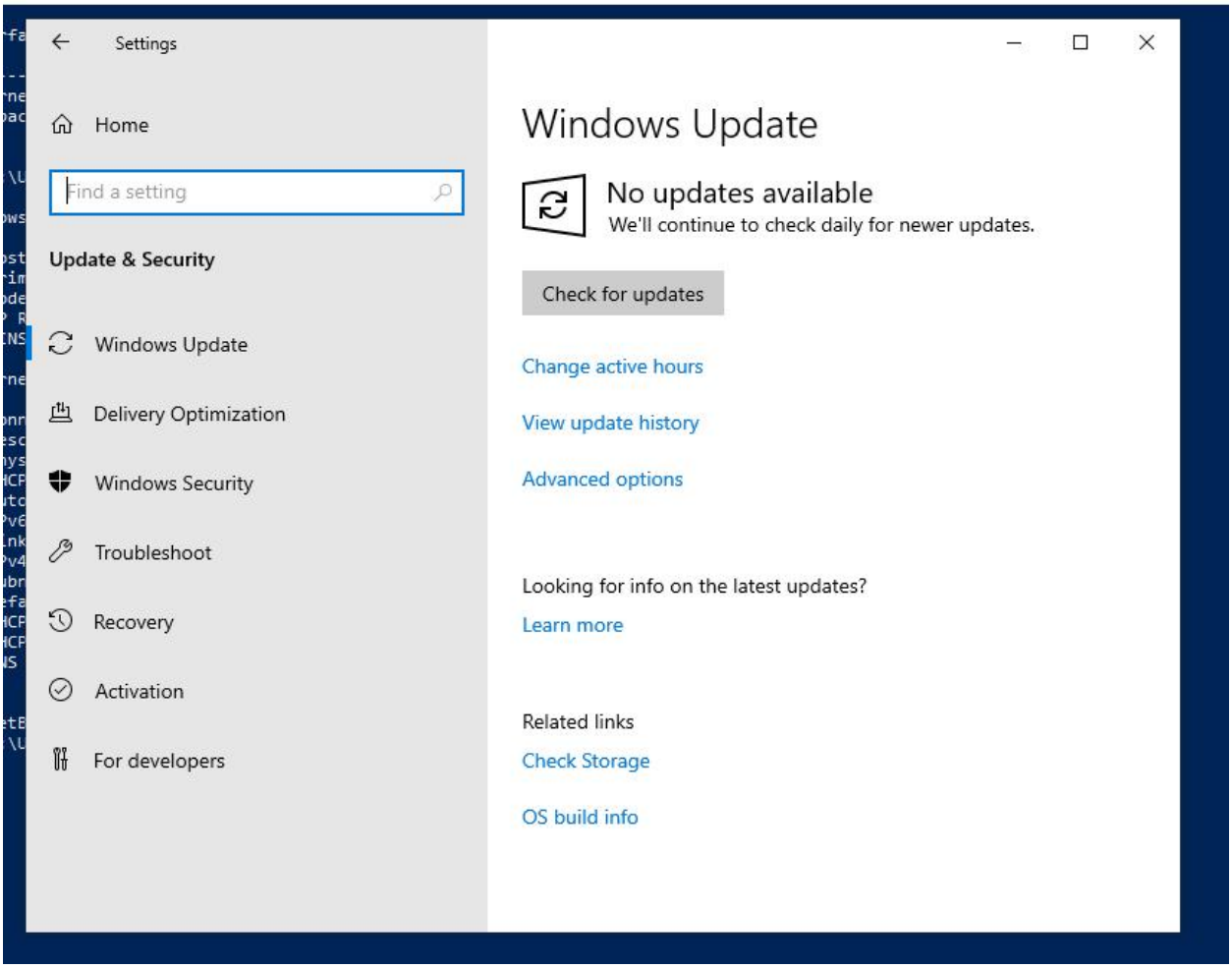
3- Configurer l'adresse IPv4

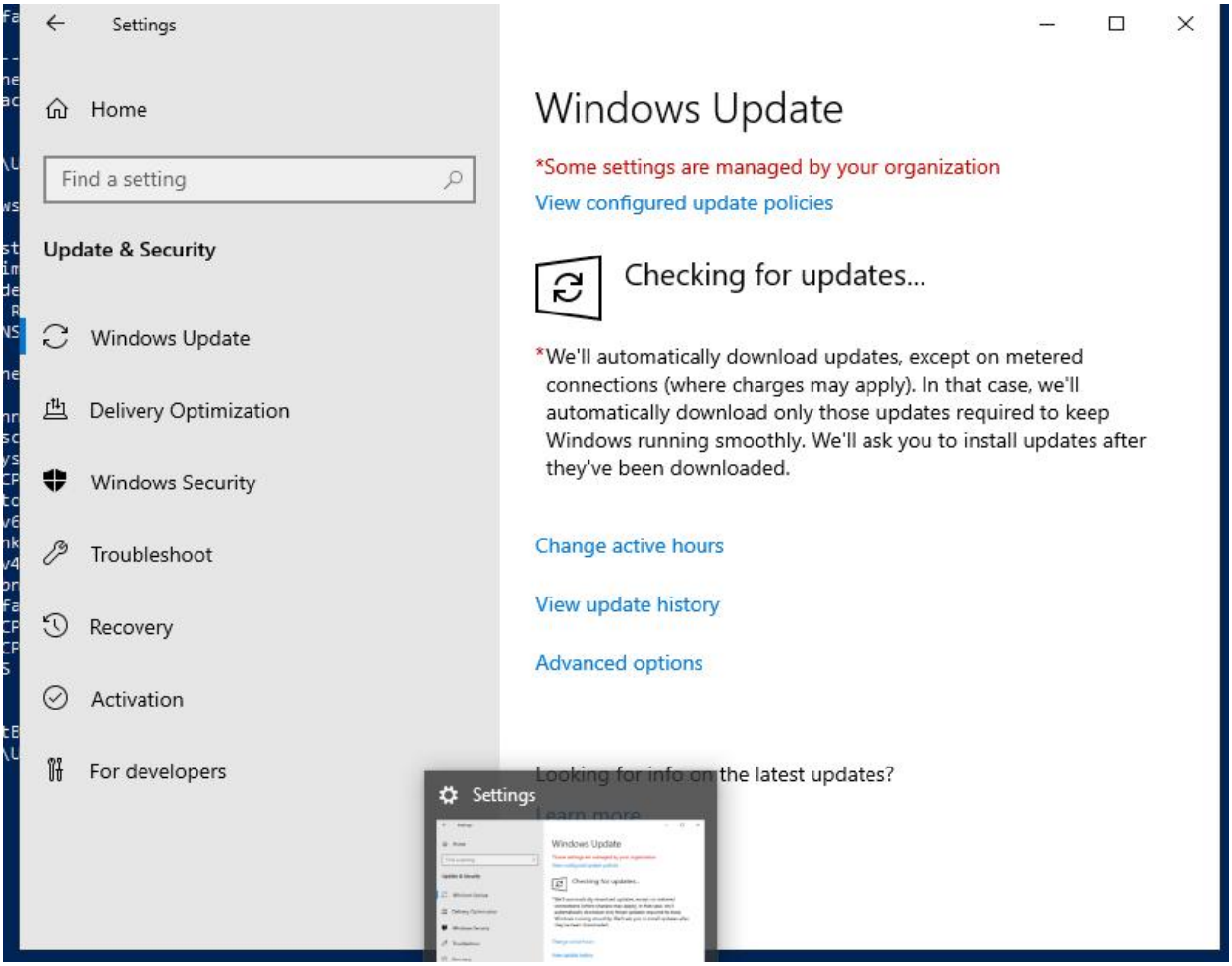

Network Connections
Network and Internet › Network Connections
Search Network Conn
Organize
Disable this network device
Diagnose this connection
Rename this connection
Ethernet Properties
Networking
Internet Protocol Version 4 (TCP/IPv4) Properties
General
You can get IP settings assigned automatically if your network supports this capability. Otherwise, you need to ask your network administrator for the appropriate IP settings.
Obtain an IP address automatically
Use the following IP address:
IP address: 192 . 168 . 10 . 10
Subnet mask: 255 . 255 . 255 . 0
Default gateway:
Obtain DNS server address automatically
Use the following DNS server addresses:
Preferred DNS server:
Alternate DNS server:
Validate settings upon exit
Advanced...
OK
Cancel
1 item
SRV-INFRA01-S 1076 Warning User32 System
SRV-INFRA01-S 1076 Warning User32 System


Administrator: Windows PowerShell
Ethernet Intel(R) PRO/1000 MT Desktop Adapter 4 Up 08-00-27-41-E1-92 1 Gbps
PS C:\Users\Administrator> Get-NetIPConfiguration
InterfaceAlias : Ethernet
InterfaceIndex : 4
InterfaceDescription : Intel(R) PRO/1000 MT Desktop Adapter
NetProfile.Name : Network 2
IPv6Address : fd17:625c:f037:2:7bb5:44ef:d87:bcb3
IPv4Address : 192.168.10.10
IPv6DefaultGateway : fe80::2
IPv4DefaultGateway :
DNSServer : fec0:0:0:ffff::1
fec0:0:0:ffff::2
fec0:0:0:ffff::3
PS C:\Users\Administrator> Get-DnsClientServerAddress -AddressFamily IPv4
InterfaceAlias Interface Address ServerAddresses
Index Family
Ethernet 4 IPv4 {}
Loopback Pseudo-Interface 1 1 IPv4 {}
PS C:\Users\Administrator> ipconfig /all
Windows IP Configuration
Host Name . . . . . : SRV-INFRA01-s
Primary Dns Suffix . . . . . :
Node Type . . . . . : Hybrid
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No
Ethernet adapter Ethernet:
Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) PRO/1000 MT Desktop Adapter
Physical Address. . . . . : 08-00-27-41-E1-92
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
IPv6 Address. . . . . : fd17:625c:f037:2:7bb5:44ef:d87:bcb3(Preferred)
Link-local IPv6 Address . . . . . : fe80::4de8:9a90:1faf:937a%4(Preferred)
IPv4 Address. . . . . : 192.168.10.10(Preferred)
Subnet Mask . . . . . : 255.255.255.0
7:11 PM
10/25/2026



4-Mettre le système à jour


Settings
Home
Find a setting
Update & Security
Windows Update
Delivery Optimization
Windows Security
Troubleshoot
Recovery
Activation
For developers
Windows Update
No updates available
We'll continue to check daily for newer updates.
Check for updates
Change active hours
View update history
Advanced options
Looking for info on the latest updates?
Learn more
Related links
Check Storage
OS build info



Questions#2

1. Pourquoi attribuer une adresse fixe à un serveur ?

Une adresse IP fixe permet au serveur d'être toujours accessible à la même adresse sur le réseau. Les postes clients et les services réseau peuvent ainsi le localiser facilement sans risque que son adresse change après un redémarrage.

2. Quelle est la fonction du masque réseau ?

Le masque réseau sert à identifier la partie « réseau » et la partie « hôte » d'une adresse IP.

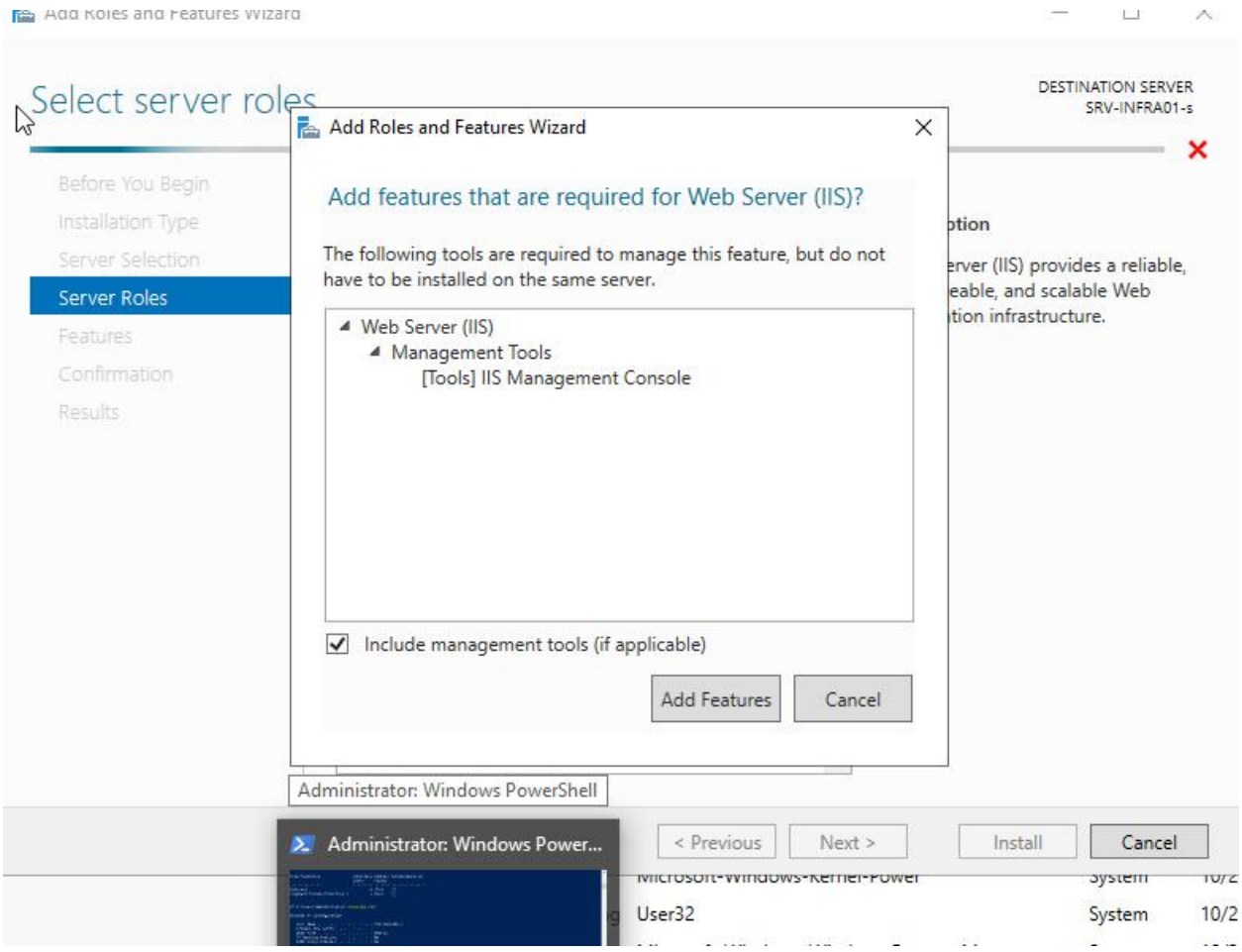
Il permet de déterminer si deux appareils appartiennent au même réseau ou s'ils doivent communiquer via une passerelle.

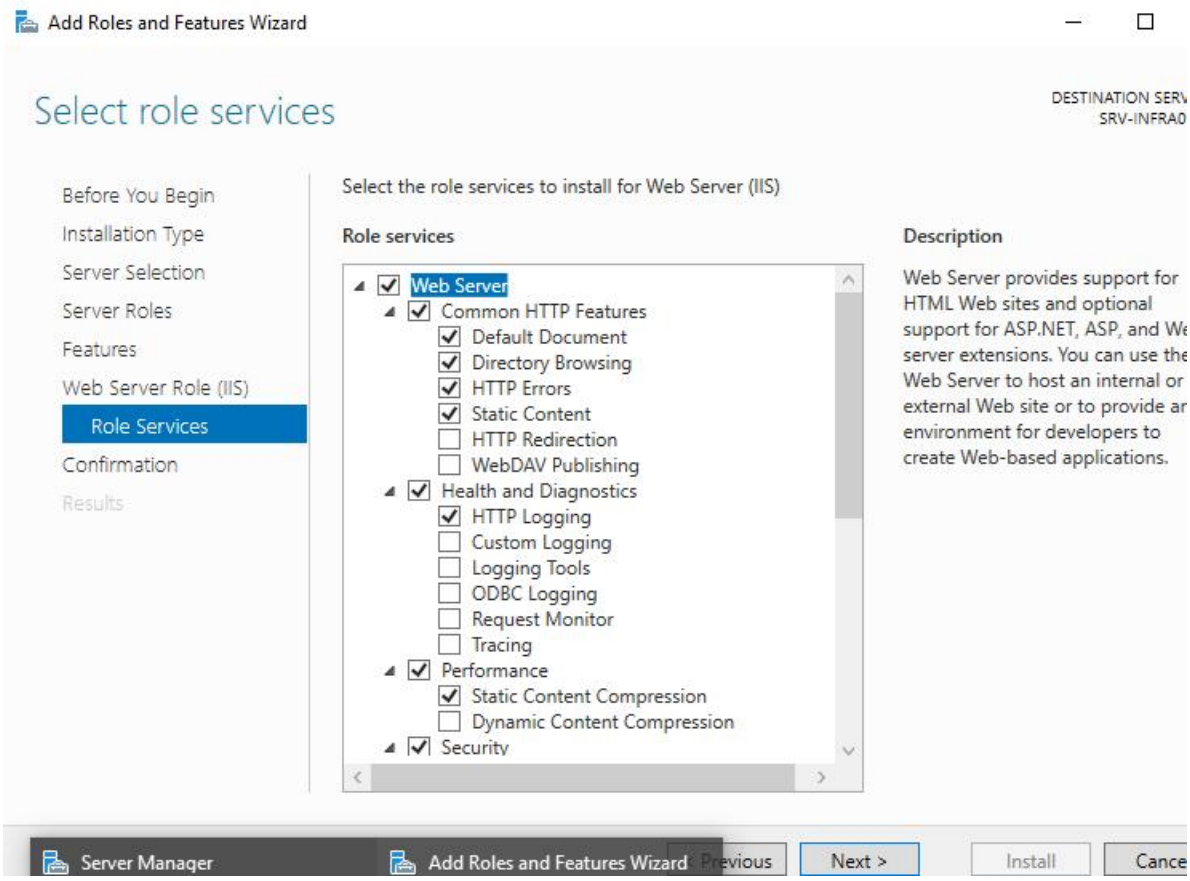
3. À quoi servent la passerelle et le DNS ?

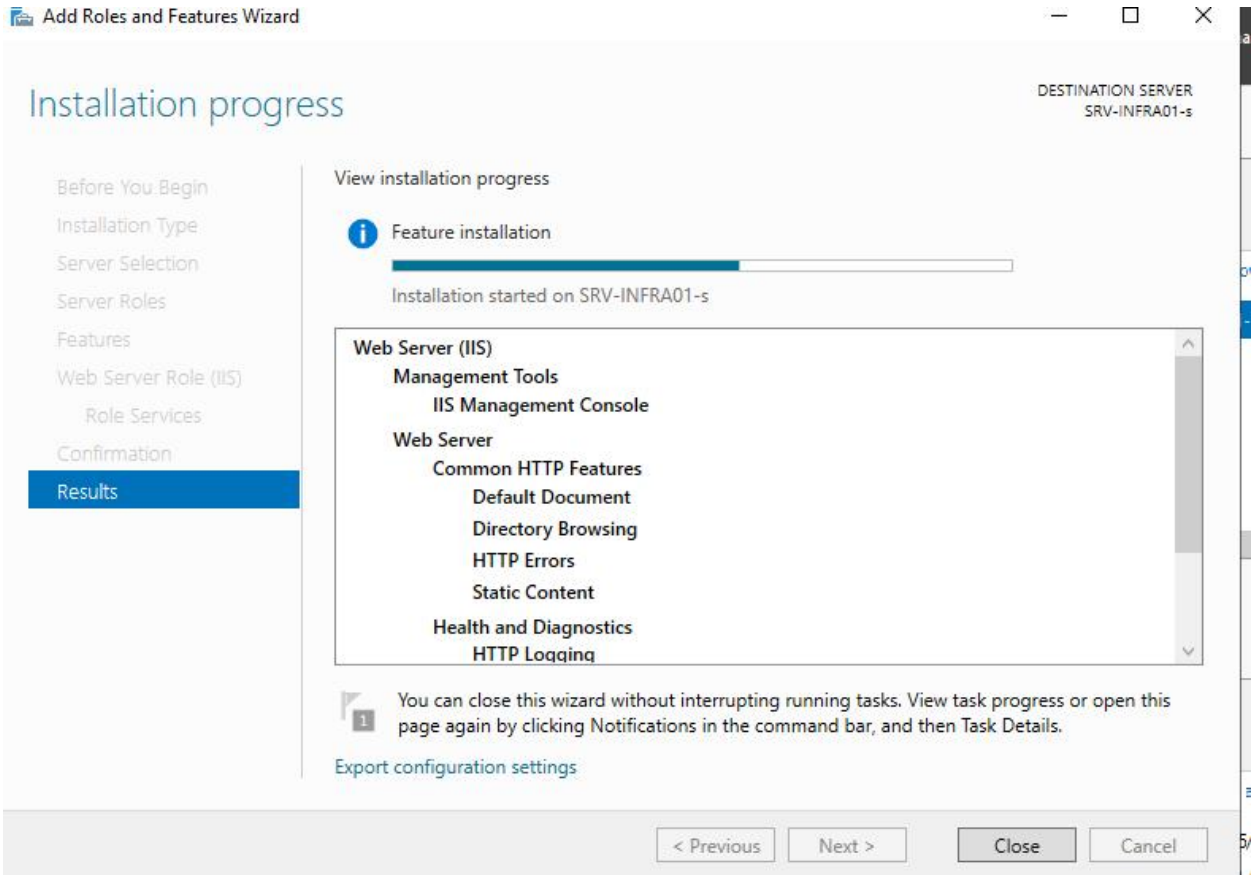
1-La passerelle (Gateway) permet aux appareils du réseau local de communiquer avec d'autres réseaux, notamment Internet.

2-Le DNS (Domain Name System) traduit les noms de domaine en adresses IP. Par exemple, il permet de convertir *www.google.com* en une adresse IP compréhensible par les ordinateurs.

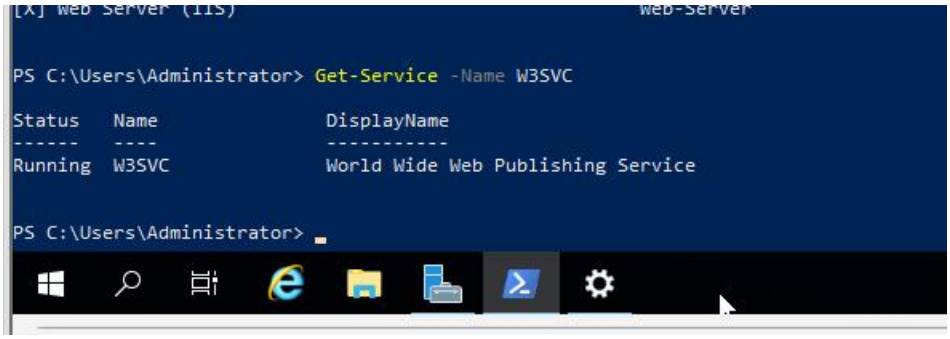
5-Installation du serveur Web IIS



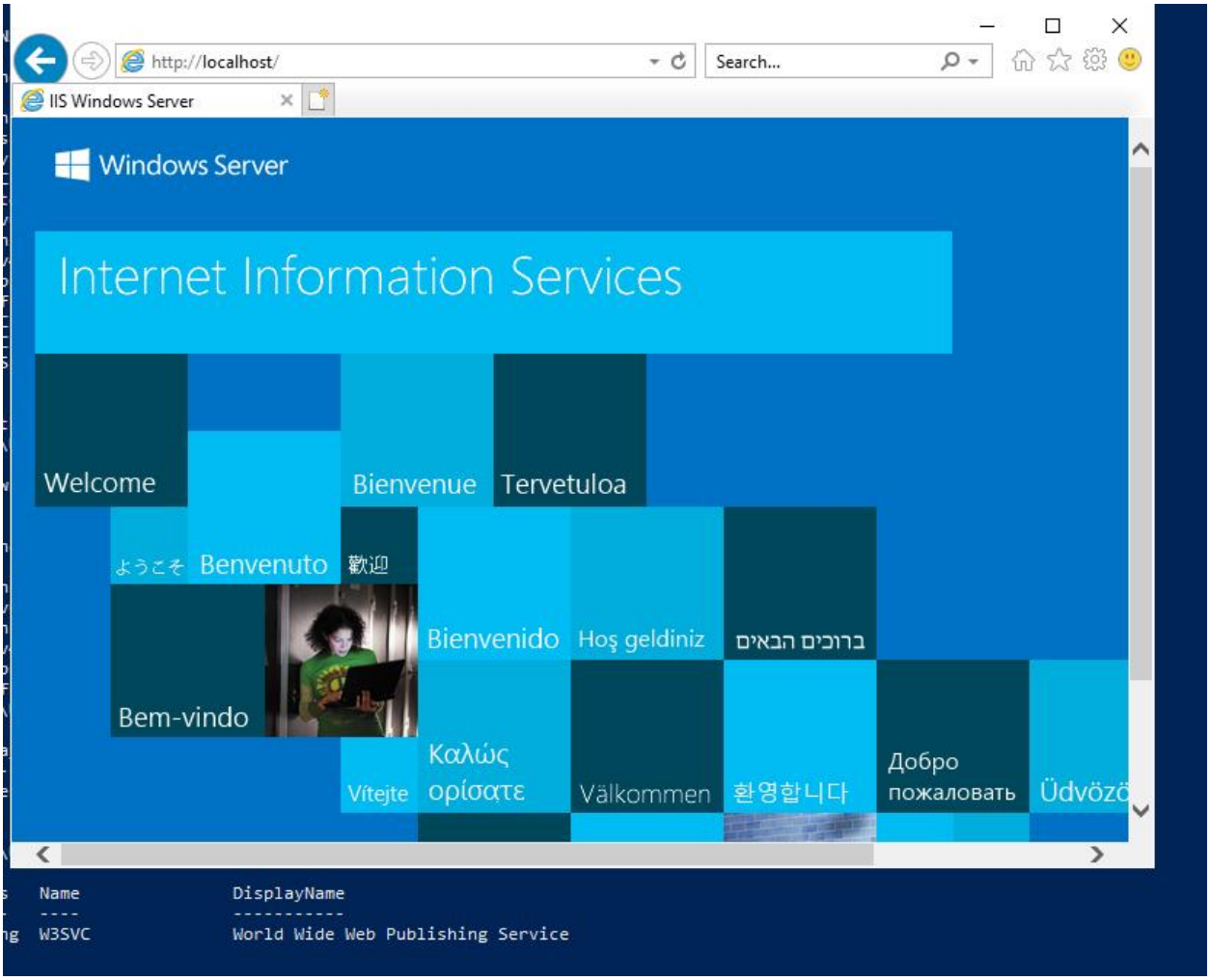

Add Roles and Features Wizard
Select role services
DESTINATION SERV
SRV-INFRA0
Before You Begin
Installation Type
Server Selection
Server Roles
Features
Web Server Role (IIS)
Role Services
Confirmation
Results
Select the role services to install for Web Server (IIS)
Role services
Web Server
Common HTTP Features
Default Document
Directory Browsing
HTTP Errors
Static Content
HTTP Redirection
WebDAV Publishing
Health and Diagnostics
HTTP Logging
Custom Logging
Logging Tools
ODBC Logging
Request Monitor
Tracing
Performance
Static Content Compression
Dynamic Content Compression
Security
Description
Web Server provides support for HTML Web sites and optional support for ASP.NET, ASP, and We server extensions. You can use the Web Server to host an internal or external Web site or to provide ar environment for developers to create Web-based applications.
Server Manager
Add Roles and Features Wizard
Previous
Next >
Install
Cance



Vérification



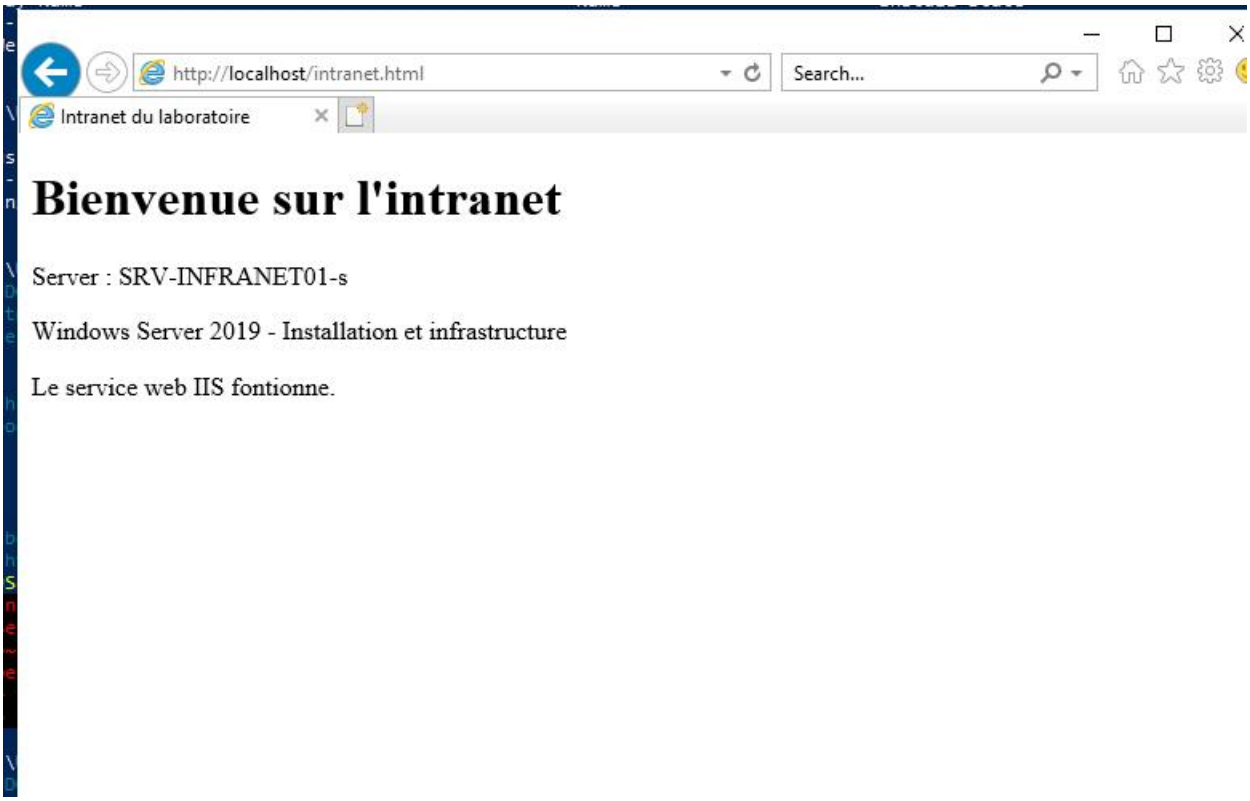
Vérifier le site par défaut



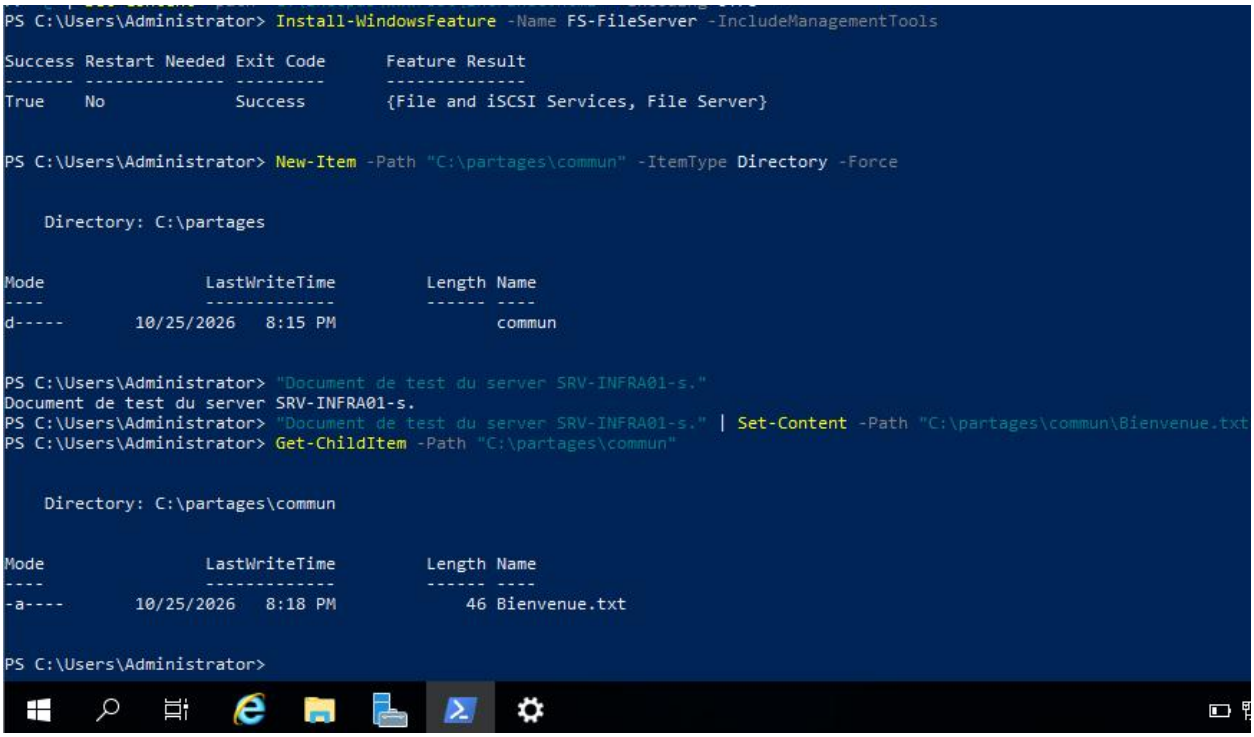
6-Créer une page intranet

```
PS C:\Users\Administrator> @'
>> <!DOCTYPE html>
>> <html lang="fr">
>> <head>
>>     <meta charset="utf-8">
>>     <title>Intranet du laboratoire</title>
>> </head>
>> <body>
>>     <h1>Bienvenue sur l'intranet</h1>
>>     <p>Server : SRV-INFRANET01-s</p>
>>     <p>Windows Server 2019 - Installation et infrastructure</p>
>>     <p>Le service web IIS fontionne.</p>
>> </body>
>> </html>
>> '@ | Set-Content -path "C:\inetpub\wwwroot\intranet.html" -Encoding UTF8
PS C:\Users\Administrator>
```

Ouvrons



7-Installation du rôle de serveur de fichiers



Questions#3

1. Quelle différence existe-t-il entre un rôle et une fonctionnalité ?

Un rôle est un service principal que le serveur fournit sur le réseau, comme le serveur de fichiers, le serveur DNS ou Active Directory. Une fonctionnalité est un composant complémentaire qui ajoute des outils ou des capacités au serveur, sans constituer un service principal.

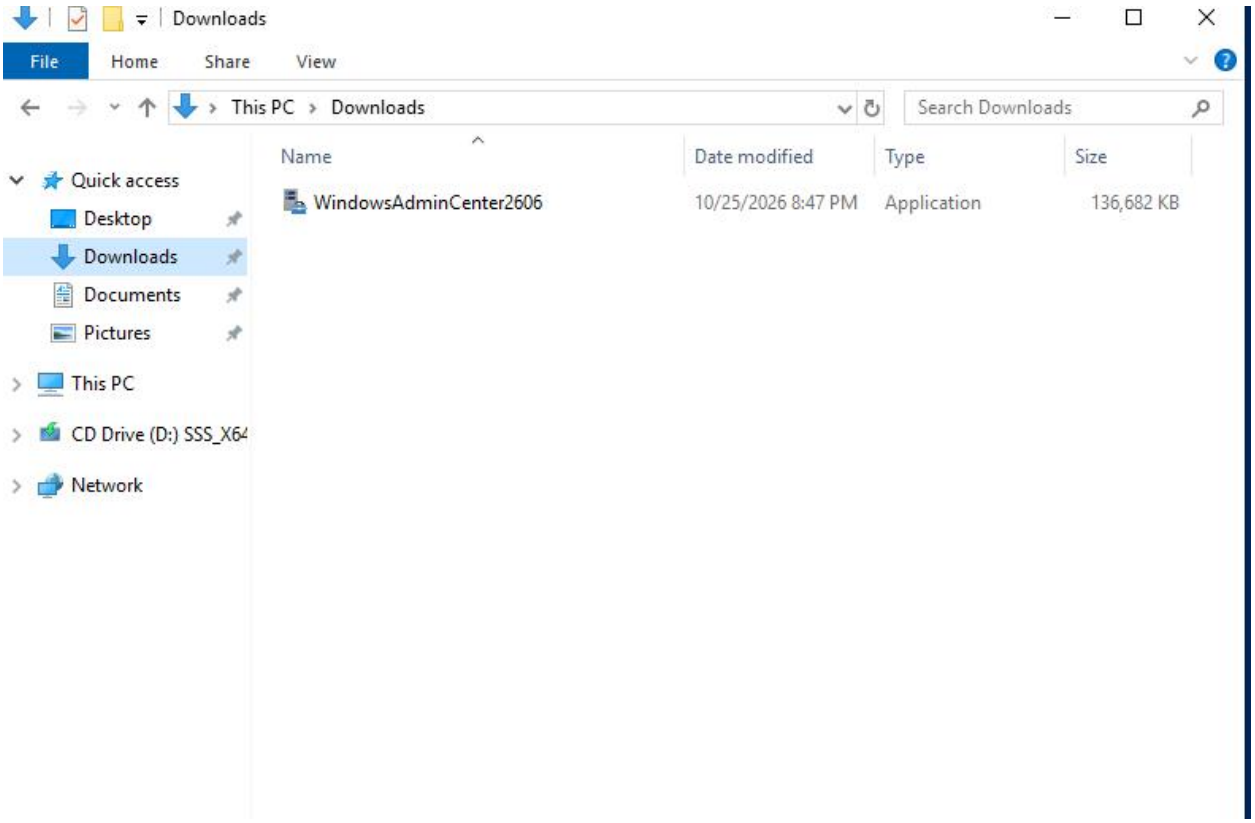
2. L'installation du rôle de serveur de fichiers crée-t-elle automatiquement un partage utilisable ?

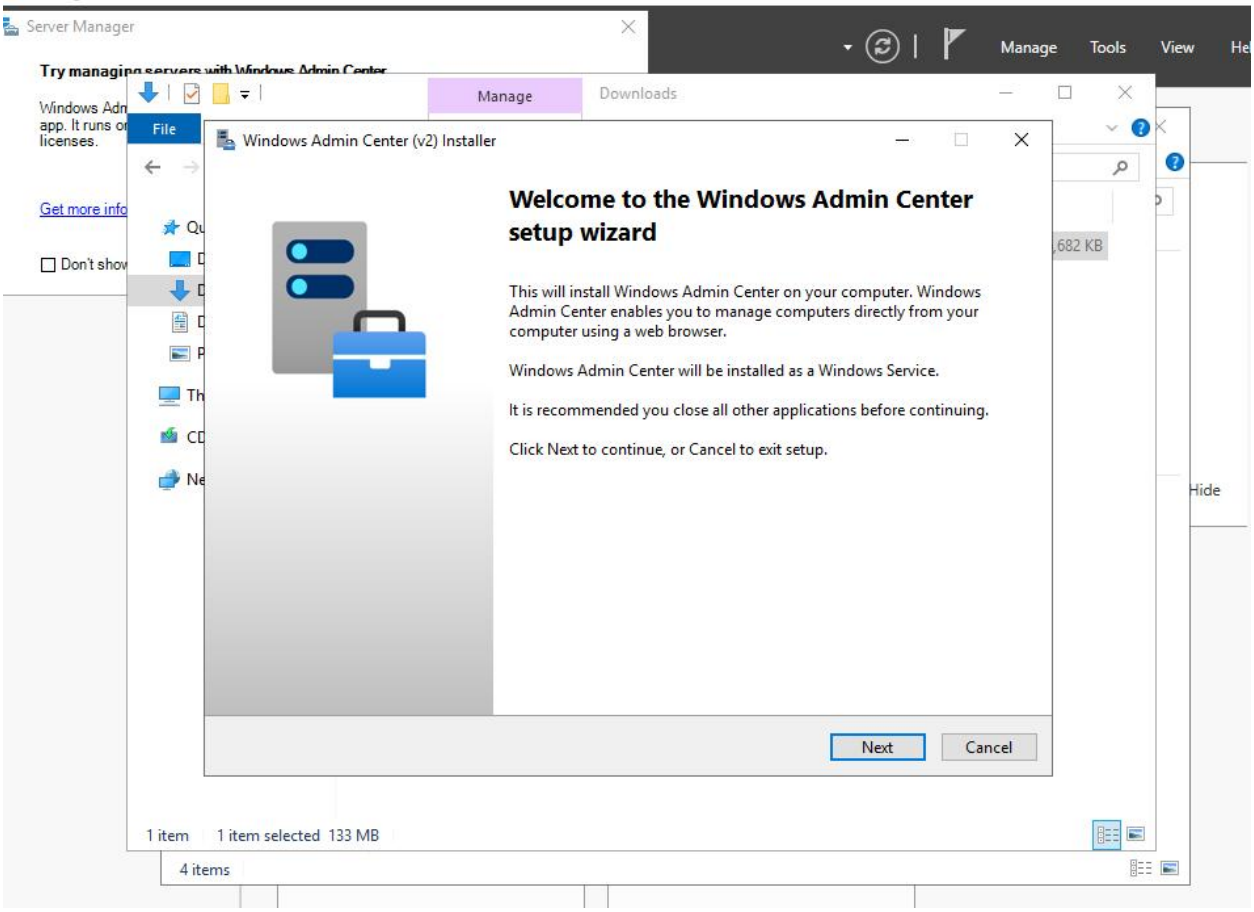
Non. L'installation du rôle Serveur de fichiers permet au serveur de gérer les partages, mais l'administrateur doit ensuite créer un dossier partagé, définir les autorisations et configurer l'accès des utilisateurs.

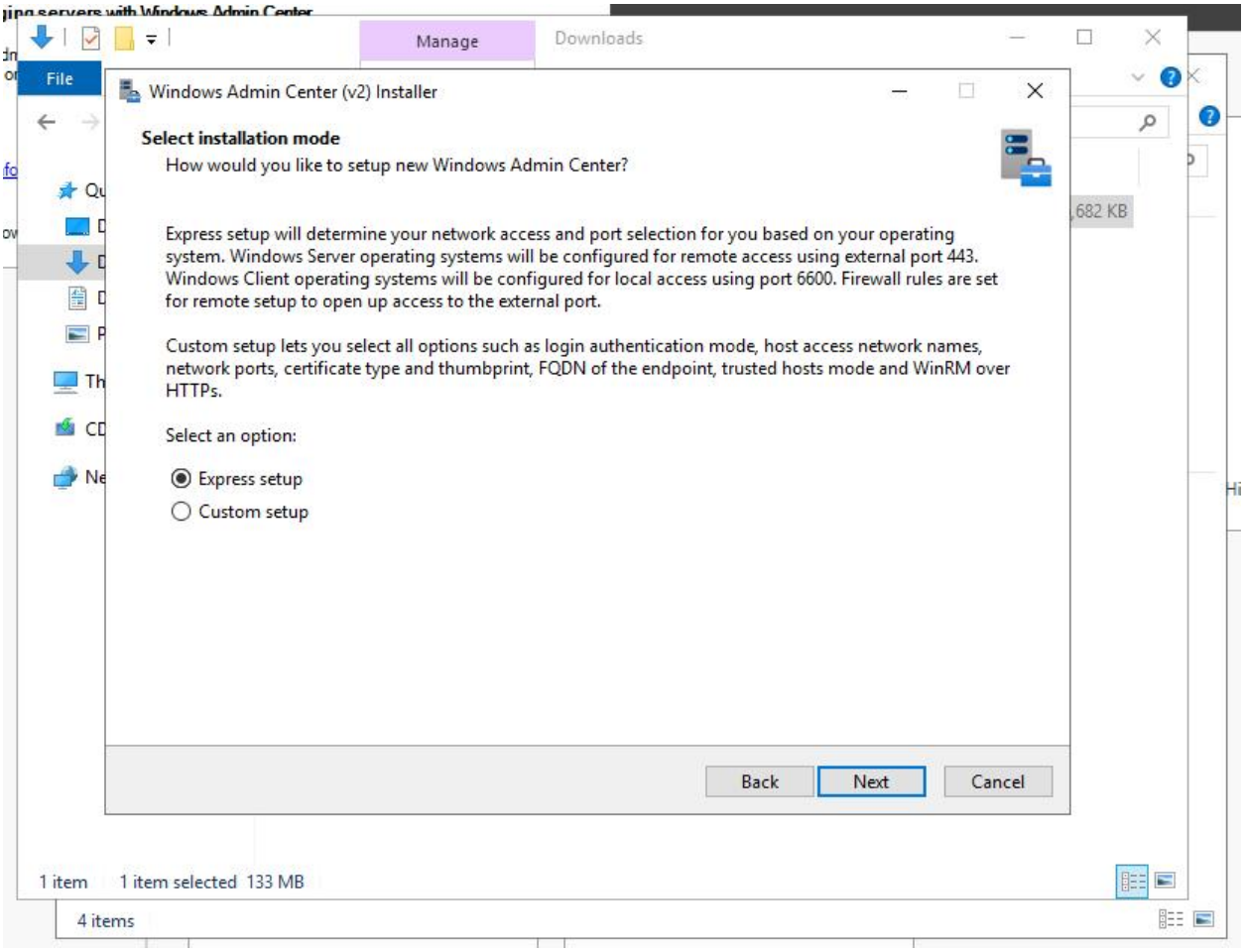
3. Quel intérêt présente PowerShell pour configurer plusieurs serveurs ?

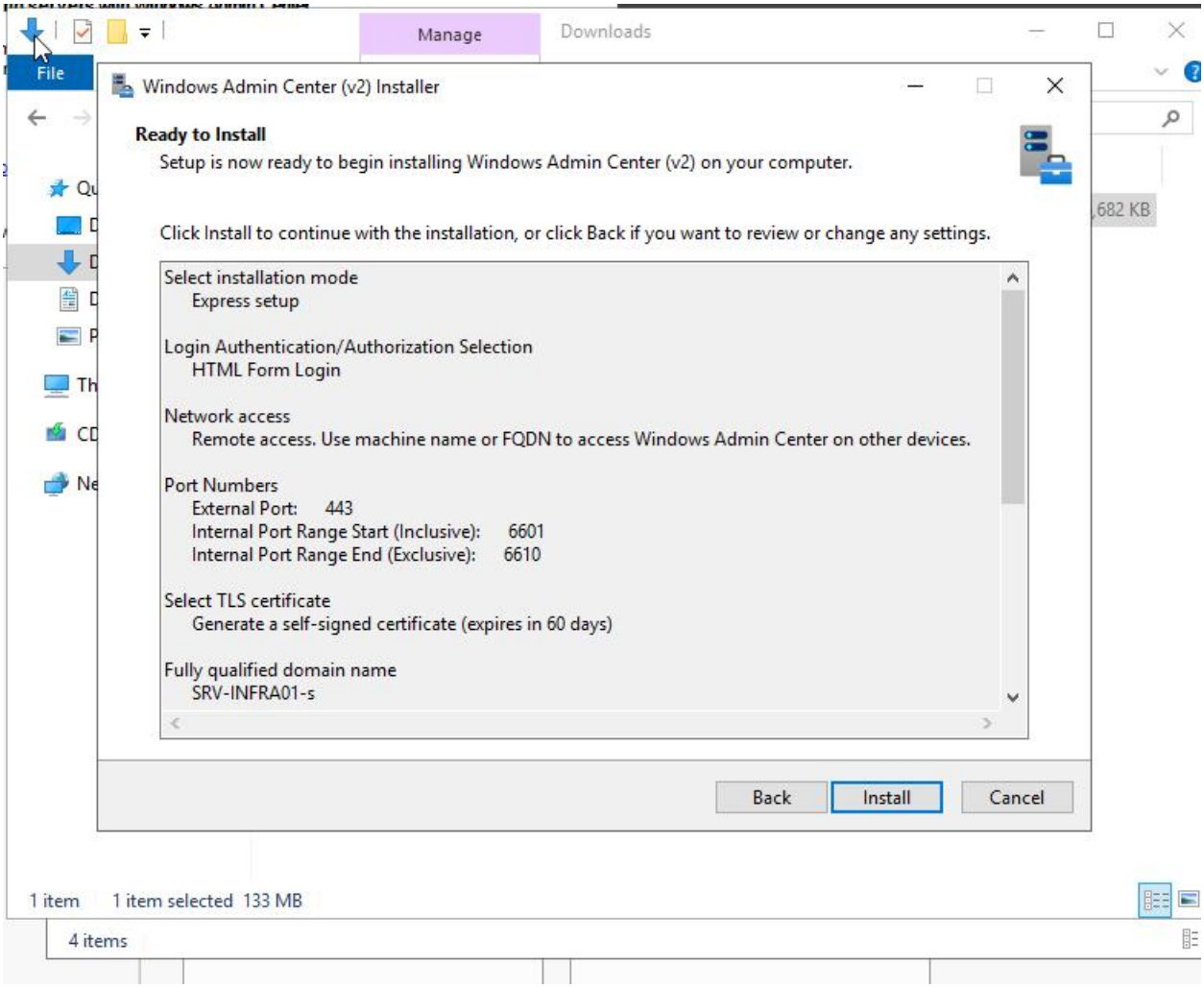
PowerShell permet d'automatiser les tâches d'administration grâce à des scripts. Il est ainsi possible d'appliquer la même configuration à plusieurs serveurs rapidement, de réduire les erreurs manuelles et de gagner du temps lors du déploiement et de la gestion des serveurs.

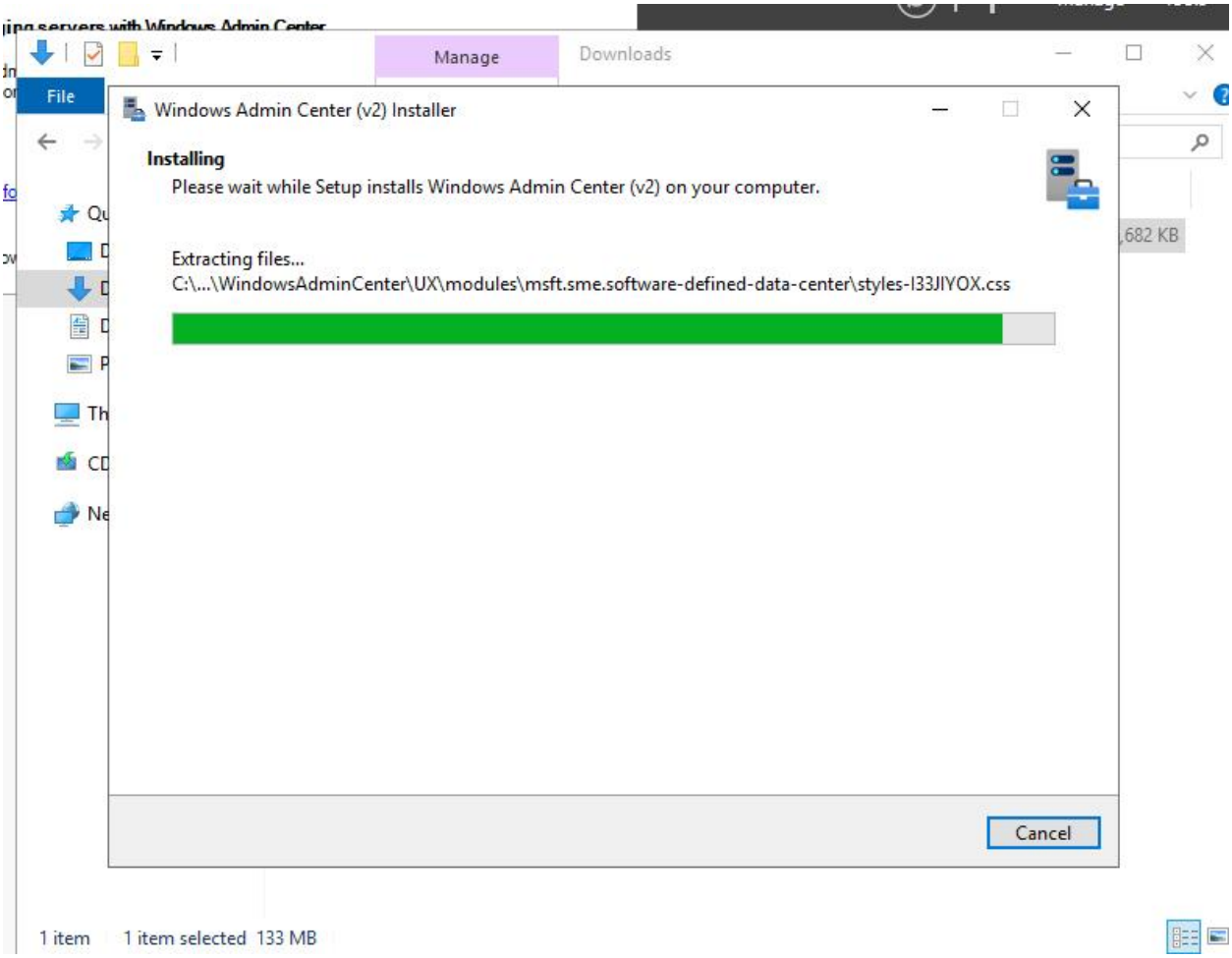
8-Installation et utilisation de Windows Admin Center

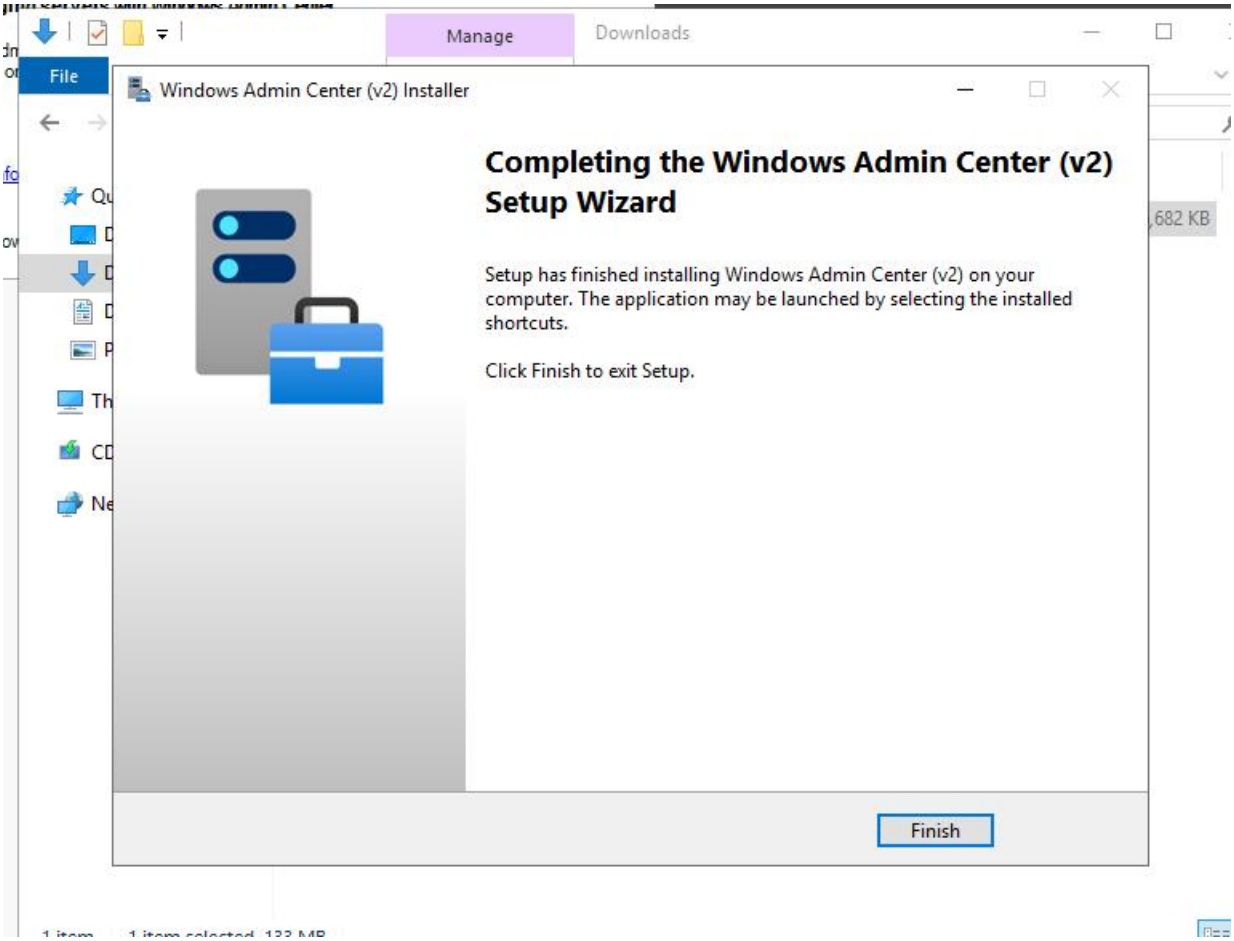

Downloads
File
Home
Share
View
This PC
Downloads
Search Downloads
Quick access
Desktop
Downloads
Documents
Pictures
This PC
CD Drive (D:) SSS_X64
Network
Name
Date modified
Type
Size
WindowsAdminCenter2606
10/25/2026 8:47 PM
Application
136,682 KB

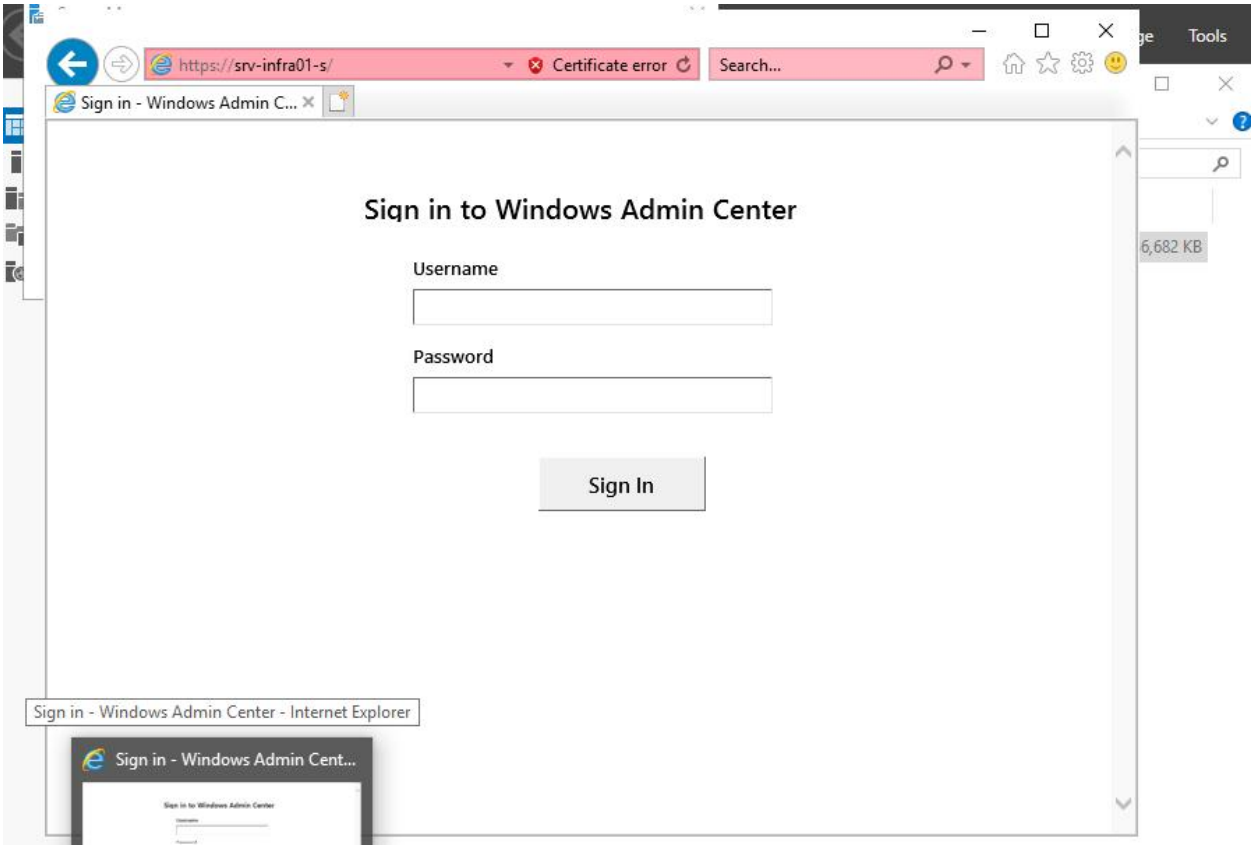

Server Manager
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Windows Admin Center (v2) Installer
Welcome to the Windows Admin Center setup wizard
This will install Windows Admin Center on your computer. Windows Admin Center enables you to manage computers directly from your computer using a web browser.
Windows Admin Center will be installed as a Windows Service.
It is recommended you close all other applications before continuing.
Click Next to continue, or Cancel to exit setup.
Next
Cancel
1 item
1 item selected 133 MB
4 items


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Windows Admin Center (v2) Installer
Select installation mode
How would you like to setup new Windows Admin Center?
Express setup will determine your network access and port selection for you based on your operating system. Windows Server operating systems will be configured for remote access using external port 443. Windows Client operating systems will be configured for local access using port 6600. Firewall rules are set for remote setup to open up access to the external port.
Custom setup lets you select all options such as login authentication mode, host access network names, network ports, certificate type and thumbprint, FQDN of the endpoint, trusted hosts mode and WinRM over HTTPs.
Select an option:
Express setup
Custom setup
Back
Next
Cancel
1 item
1 item selected 133 MB
4 items


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Windows Admin Center (v2) Installer
Ready to Install
Setup is now ready to begin installing Windows Admin Center (v2) on your computer.
Click Install to continue with the installation, or click Back if you want to review or change any settings.
Select installation mode
Express setup
Login Authentication/Authorization Selection
HTML Form Login
Network access
Remote access. Use machine name or FQDN to access Windows Admin Center on other devices.
Port Numbers
External Port: 443
Internal Port Range Start (Inclusive): 6601
Internal Port Range End (Exclusive): 6610
Select TLS certificate
Generate a self-signed certificate (expires in 60 days)
Fully qualified domain name
SRV-INFRA01-s
Back
Install
Cancel
1 item
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4 items


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Windows Admin Center (v2) Installer
Installing
Please wait while Setup installs Windows Admin Center (v2) on your computer.
Extracting files...
C:\...\WindowsAdminCenter\UX\modules\msft.sme.software-defined-data-center\styles-I33JIYOX.css
Cancel
1 item
1 item selected 133 MB


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Windows Admin Center (v2) Installer
Completing the Windows Admin Center (v2) Setup Wizard
Setup has finished installing Windows Admin Center (v2) on your computer. The application may be launched by selecting the installed shortcuts.
Click Finish to exit Setup.
Finish


Tools
https://srv-infra01-s/
Certificate error
Search...
Sign in - Windows Admin C...
Sign in to Windows Admin Center
Username
Password
Sign In
6,682 KB
Sign in - Windows Admin Center - Internet Explorer
Sign in - Windows Admin Cent...